



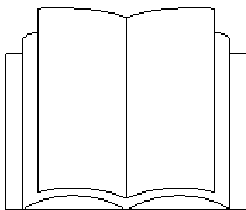
Annex for the original Instruction Manual



Log Cutter/Splitter

JAPA 305

Product number:	300-700-series loaders:	A415435
	200-series loaders:	A415434



Please read carefully the original attachment's instruction manual before using the equipment, as well as this annex, and follow all instructions.

This annex for the operator's manual includes specific information about the use of the equipment with Avant multi purpose loaders.

Keep the original operator's manual as well as this annex for later reference.

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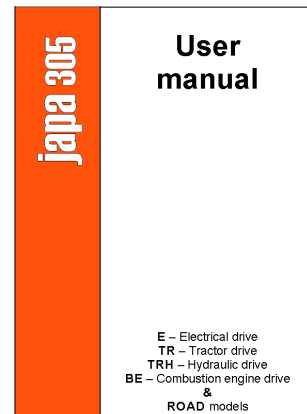
CONTENTS

1. Foreword	4
2. Designed purpose of use.....	5
3. Safety instructions for using the log splitter with Avant loaders.....	5
4. Technical specifications.....	7
5. Assembly of the log cutter/splitter	8
5.1. Mounting the log cutter/splitter to the loader	8
5.2. Connecting and disconnecting the hydraulic couplings	9
6. Operation	10
6.1. Setting up the log cutter/splitter	11
6.2. Use	11
7. Maintenance and warranty	12

Annex: JAPA 305 User Manual



This annex for the original User Manual of the log cutter-splitter does include all necessary information about safe and correct use of the equipment. Rnsure that you have received and read also the original User Manual, which is normally shipped with the machine. Contact your sales representative or the manufacturer immediately, if the original User Manual is not available.



1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for AVANT multi purpose loaders. The log cutter/splitter has been adapted for use with AVANT loaders by the original manufacturer of the equipment. The original User Manual, provided with the log splitter/cutter, includes important information about safe use of the equipment. This annex for the manual includes important information of coupling and using the equipment with AVANT loaders.

By familiarising yourself with and following the instructions shown in the original User Manual and the additional information in this annex, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Ensure that all persons using the loader and the log cutter/splitter have received proper guidance and familiarised themselves with the manuals of the loader and attachment and all safety instructions before using the equipment.

Keep this annex of the manual and the original user manual at hand throughout the service life of the equipment. If this annex for the manual or the original operator's manual is lost or damaged, you can request a new one from your nearest AVANT dealer. If you sell or transfer the equipment, be sure to hand both parts of the manual to the new owner.

This manual uses the following warning symbols to indicate important points that must be taken into account to reduce the risk of personal injury or property damage:

	WARNING: This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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Note:

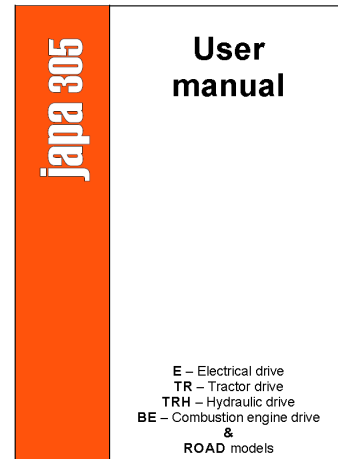
This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This annex for the instruction manual is original English version.



This annex for the original instruction manual of the log cutter/splitter machine does not cover all necessary information needed for safe operation of the equipment. This annex is not intended to be distributed separately. The original User Manual, provided with the machine, should be considered as the main source of information about safety, use, technical information, and the maintenance of the equipment. This annex includes important information about the use of the equipment specifically with AVANT multi purpose loaders.



2. Designed purpose of use

The JAPA 305 Avant model firewood machines are hydraulically operated attachments designed and manufactured to be suitable for use with 200-700 –series AVANT multi purpose loaders. They are intended to be used by one person at a time for cutting firewood and splitting it into halves or quarters. The log cutter/splitter cuts the log with a chain saw blade to the adjusted length, and then the log is split with a hydraulic cylinder, which presses the log against a splitting wedge. Split logs will get pushed automatically out, or the optional conveyor system can be used.

The log cutter/splitter is intended to be used only when it is securely locked to the quick coupling plate of AVANT loader. The attachment is powered by the hydraulic outlet of the Avant loader. The log conveyor, shown in the original manual, is optional extra. Contact your AVANT dealer for further information about its availability.

Familiarise yourself with the instruction manual's instructions regarding service and maintenance. Demanding repair and maintenance operations, such as hydraulic repairs, should be left for professional maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for using the log splitter with Avant loaders

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



Misuse or careless use, or using equipment that is in bad condition, may cause serious injuries. Get to know the controls of the loader and coupling procedures of the attachment at a safe area. In particular, familiarize yourself with procedures how to stop the equipment in a safe manner and to make it safe. Read the following safety precautions carefully.



- When attaching the attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are in the lower position and that they have locked the attachment to the loader.**
- The log cutter/splitter is to be used by one operator at a time. Keep others at a safe distance, and do not let others to operate the controls of the loader while using the attachment.
- In case of fault or blockage, make sure that the equipment has been made safe by turning off the engine of the loader. When a blockage is released the moving parts of the attachment may move unexpectedly, causing the saw chain to move or pushing of the splitting cylinder. Follow the safe stopping procedure before handling the chainsaw or when reaching to the crushing hazard area of the splitting cylinder.
- When transporting the equipment, pay attention to the contours of the terrain as well as other hazards, such as loose rocks or branches and trees reaching the driver's area.
- Always transport the log cutter/splitter as low and close to the machine as possible to keep the centre of gravity low. Keep the telescopic boom retracted and drive calmly.
- Keep hands and feet away from moving parts.
- When uncoupling the log cutter/splitter from the loader, ensure that it is secured against accidental movement.
- Never perform any maintenance or adjustments while the loader engine is running. Also, release the residual hydraulic pressure of the system.
- Do not modify the structure of the attachment in a manner that would affect its safety. Check that all safety features shown in the original User manual are in working condition.
- Also read the safety instructions for the loader in the loader manual. Get to know the loader controls and how to operate it safely.

Safe stopping of the equipment, before going near the blade or splitting area:



- Stop the attachment by turning its control lever to position "off" and turn off auxiliary hydraulics of the loader.
- Make sure that the log machine is firmly against the ground and will remain stable.
- Shut down the loader engine and lock the parking brake.
- Release residual pressure from the hydraulic system by moving the auxiliary hydraulics control lever to its extreme positions.
- Remove key from ignition to make sure that another person can't operate the controls of the loader.



Note that the loader boom moves down and the log machine can be tilted even if the engine has been stopped. Never go under or in front of raised boom or attachment. Always lower the attachment on the ground before leaving the driver's seat.



Note the effect of heavy load to the the stability and handling characteristics and of the loader, especially when operating with the smaller loader models. Always keep the load close to the ground and keep the telescopic boom retracted. Drive slowly and carefully and avoid steep slopes. Use additional counterweights as needed. Read the operator's manual of the loader.



Remember to wear proper protective clothing:



- The noise level at the operator's area may exceed 85 dB(A). Wear hearing protection.
- Wear safety boots while working with the loader.



- Wear protective gloves. and when handling hydraulic. Wear also safety goggles when handling hydraulic hoses or components.

4. Technical specifications

The labels and markings, which must be visible on the equipment, are listed in the original User Manual. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer.

Table 1 –Log cutter/splitter specifications

Product number	A415434	A415435
Log length, min-max:	200 - 500 mm	
Log diameter, min-max:	20 - 300 mm	
Weight:	265 kg	
Minimum operating temperature:	-25°C	
Saw flange:	13"	
Splitting force:	30 kN	
Dimensions, transport configuration: L x W x H:	2780 x 1030 x 2450 mm	
Maximum input of hydraulic energy:	30 l/min, 22,5 MPa (225 bar)	
Suitable AVANT loader:	216, 218, 220	320-750

5. Assembly of the log cutter/splitter

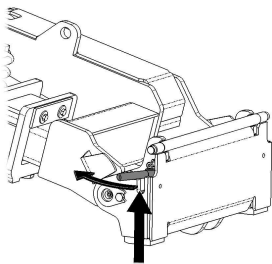
The log cutter/splitter is ready for use as delivered. Attach the attachment and its hydraulic hoses according to the following instructions.

5.1. Coupling the log cutter/splitter to the loader

The attachment is mounted with the quick attach plate of the loader boom and the counterpart on the attachment, which are not shown in the original User Manual. The locking is semi-automatic, but due to the relatively high position of the quick attach plate on the log splitter, and the risk of tipping over of the attachment, the locking pins must always be locked manually.

Coupling the attachment to the loader is quick and easy, but it must be done carefully. The attachment is fixed to the loader boom by using the quick attach plate and the counterpart on the attachment. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must not be raised over one metre under any circumstance if it has not been locked. To prevent hazardous situations, always follow the attachment coupling instructions provided in the following pages. Also remember the safety instructions shown in Chapter 3 and additional information shown in the operator's manual of the loader. The attachment is mounted to the loader as follows:

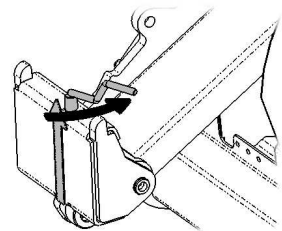
300-700 series loaders:



Step 1:

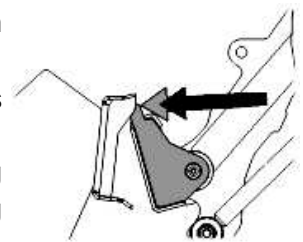
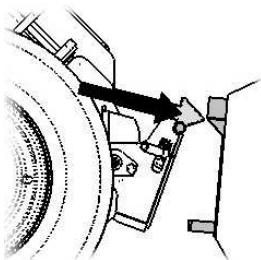
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.

200 series loaders:



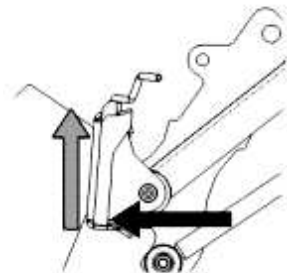
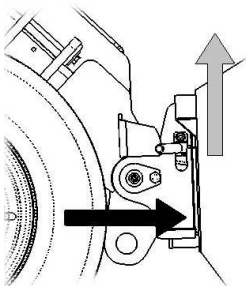
Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this. Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.



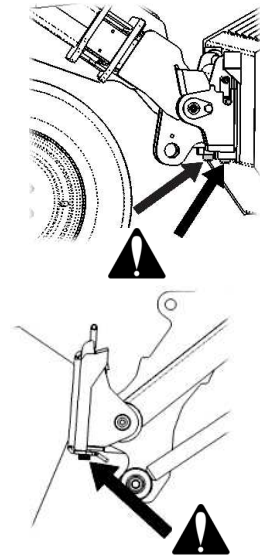
Step 3:

- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment. Lock the
- Lock the locking pins manually or lock the hydraulic locking. Always check the locking of pin / pins.





An attachment that has not been completely locked to the loader may fall on the boom or operator, or under the loader during driving, and cause serious injuries or loss of control of the loader. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or lift an attachment that has not been locked.



Do not use the automatic locking of the locking pins with the log cutter/splitter, because the quick attach plate of the machine is at high level, and tilting of an unlocked log cutter/splitter increases the risk of toppling over. It is recommended to lower the attachment on the ground and secure the locking manually, as it is necessary to leave the driver's seat to connect the hydraulic couplings.

5.2. Connecting and disconnecting the hydraulic fittings

On 300-700 series loaders hydraulic hoses are connected using the multiconnector system. . If you want to change the type of the hydraulic couplings, contact your Avant dealer or service point for instructions or installation services. On 200-series loaders only conventional quick couplings are available.



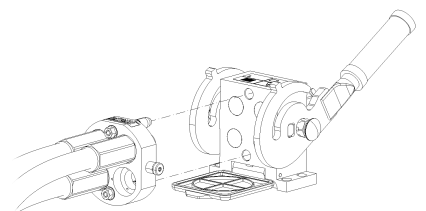
Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries or burns.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



Connecting conventional quick couplings:

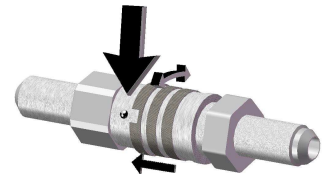
Before connecting or disconnecting the standard quick couplings, the residual pressure must be released as shown below. The conventional quick couplings will not connect, if there is pressure in the hydraulic system.

To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Make sure that both fittings are properly connected, otherwise the hydraulic motor of the mower may get damaged.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. There is a small groove in the collar of all female fittings, and a small ball near the collar. When connecting or disconnecting, the groove and ball must be aligned. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The collar may also turn accidentally to the locking position, making it impossible to connect or disconnect the couplings.



Releasing residual hydraulic pressure:

To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader to its extreme positions before disconnecting the couplings. If the fittings will not connect to the loader, the residual pressure must be released. The residual pressure must be released before handling the blades of the attachment and before storage of the attachment.

Disconnecting the hydraulics:

Before disconnecting the fittings, place the attachment on solid and level surface, as described in following paragraph. Turn the control lever of the auxiliary hydraulics to its neutral position.



When uncoupling the attachment from the loader, always disconnect the hydraulic fittings first before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent dirt from entering the hydraulic system.

6. Instructions for use

Once the log cutter/splitter and its hydraulic hoses have been coupled according to the instructions, it is ready for work. See the original operator's manual of the equipment for more information about operating the machine as well as the following additional information.



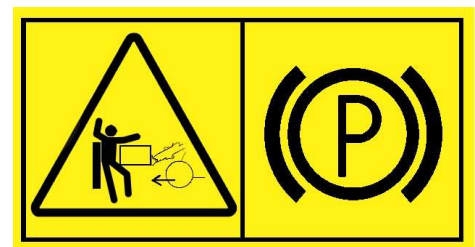
In addition to this annex of the manual, also read the original User manual for correct adjustments, operation and maintenance of the equipment, or cleaning of the equipment if the splitter can start by accident. Shut down the loader engine first.



Never perform any adjustments, maintenance or cleaning of the equipment if the splitter can start by accident. Follow safe stopping procedure to ensure that the machine can't start by accident.

6.1. Setting up the log cutter/splitter

- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. The equipment must not be used, if there is a fault in the hydraulic system of the loader or the attachment. Keeping the splitter clean will reduce the possibility of blockages.
- Transport the log cutter/splitter carefully and note the effect of weight to the stability of the loader.
- Choose a level working site, which can carry the load of the equipment and the logs.
- The operating area must be even and it must not be slippery. Clear the area from any obstacles that could cause a tripping hazard.
- Operate the log cutter/splitter in daylight or outdoor area with good artificial light.
- Check regularly to ensure there are no hydraulic leaks. Refer to original instruction manual for maintenance instructions.
- Ensure that the loader and the attachment are used in a safe and intended manner. Smoking near the operating area is prohibited. Do not allow children to operate the equipment.
- Make sure to lower the attachment securely on the ground. Ensure that the parking brake of the loader is engaged and that the splitter can not move during use.



6.2. Use



Start the machine only after it has been set to working position and work can be started. Always turn off the auxiliary hydraulics when the attachment is not in use. Do not leave the machine unattended when the engine of the loader is running. Follow safe stopping procedure always, when handling the chainsaw or when reaching the splitting area is necessary for cleaning, splitter wedge height adjustment or when removing blockages. The attachment is intended to be used by only one person at a time.



When operating the log splitter the loader must be equipped with a hydraulic oil cooler. Without the cooler hydraulic oil may overheat. Overheating of the oil decreases power and affects the service life of the hydraulic components of both the attachment and the loader. Clean the loader oil cooler core regularly; refer to the loader manual for more instructions.

To start the log cutter/splitter, turn the auxiliary hydraulics lever on the loader to its locked position and set moderate engine rpm, depending on the working pace. Depending on the loader model, two hydraulic pumps may be used to reduce engine noise. Do not exceed the maximum permissible input of hydraulic oil; full rpm must not be used when operating with 600 or 700 series loaders. Note that unnecessarily high engine rotating speed will not improve the splitting force. Operate the log cutter/splitter and perform maintenance procedures as instructed in the original manual of the equipment, so that the performance of the machine will remain at high level.



Check the operation and the adjustments of the splitter with a few empty cycles before loading a log.

The lubrication of the log cutter chain saw consumes a small amount of hydraulic oil from the loader. Check the hydraulic oil level regularly.



Do not use the attachment for any other purpose than for cutting and splitting firewood.

7. Maintenance and warranty

Instructions for maintenance and service are provided with the original instruction manual as well as the spare parts list. Contact your AVANT dealer for parts and service.

The manufacturer of the log cutter/splitter grants the warranty of one year for the log cutter/splitter. Warranty terms are provided with the instruction manual. For warranty claims, contact your AVANT dealer for further information.

The manufacturer of the log cutter/splitter provides the declaration of conformity for the equipment.

japa 305

User manual

E – Electrical drive
TR – Tractor drive
TRH – Hydraulic drive
BE – Combustion engine drive
&
ROAD models

TABLE OF CONTENTS

1. Introduction	2
1.1 Customer registration	2
1.2 Declaration of conformity	3
1.3 Purpose of use	4
1.4 Instruction and warning labels on the machine	4
1.5 The machine's type plate	5
1.6 Machine models	2
1.7 Safety instructions	2
1.8 Noise level and vibration	3
1.9 Guarantee terms	3
2. Installation of the machine	5
2.1 Delivery inspection	5
2.2 Main components of the machine	5
2.3 Lifting and transporting the machine	6
2.4 Working position	7
2.5 Connecting the power source	9
2.6 Transport position	9
3. Operating the machine	10
3.1 Acknowledging the hazards and responsibilities involved	10
3.2 Before use	10
3.3 Safety mechanism	10
3.4 Cutting device	11
3.5 Splitting device	12

4. Maintenance and troubleshooting, all models	13
4.1 Maintenance table.....	13
4.2 First maintenance	14
4.3 Daily maintenance (10 h).....	14
4.3.1 Maintaining the cutting blade	15
4.3.2 Maintaining the splitting blade.....	16
4.4 Monthly maintenance (200 h)	16
4.4.1 Adjustments of the cutting device	17
4.4.2 Adjustments of the splitting mechanism.....	18
4.4.3 Adjustment of the acceleration valve	18
4.4.4 Adjusting the conveyor's relief valve.....	18
4.5 Annual maintenance (1,000 h).....	19
4.5.1 Extended maintenance interval.....	20
4.5.2 Lubricant table.....	20
4.6 Storage	20
4.7 Maintenance log.....	20
5. Troubleshooting.....	21
6. Technical specifications	22
6.1 Technical specifications of the combustion engine (BE models)	22
6.2 Electric motor connection diagram (E models).....	23
6.3 Hydraulics diagram	24
6.4 Blade lubrication diagram	26
6.5 Safety instructions for transport on public roads (ROAD models)	27
6.6 Spare parts	28

1. Introduction

Laitilan Rautarakenne Oy (JAPA) is a Finnish company that engages in extensive product development. Our goal is to produce simple, reliable and structurally durable machines with a long service life. If you use your JAPA machine correctly and maintain it according to the instructions in this manual, it will serve you efficiently for a long time. If you have questions and would like to discuss them in detail, please contact our retailer or us directly.

CONGRATULATIONS ON PURCHASING A NEW JAPA FIREWOOD MACHINE!

This manual is intended for operators with the appropriate expertise. For this reason, operators are expected to have general knowledge and skills for work with this kind of machinery. Familiarise yourself with the manual before installing the machine and beginning operation.

Carefully study the machine's features and safety equipment before operation. Always keep this manual with the machine. The instructions, descriptions and technical specifications in this manual are based on the latest information with regard to the workings of the machine at time of printing. However, the manufacturer develops and updates the machine continuously and therefore reserves the right to change the machine's features and safety-related properties without separate notification.

In order to receive quick and efficient assistance when ordering spare parts and encountering any possible fault situations, you must indicate the information on the machine's type plate to the retailer or service technician. Write down the type plate information in the reserved section on this page to ensure that they are readily available at all times. If you cannot find a solution to a problem on your own, please contact the retailer, who will then resolve the matter with the manufacturer.

We at JAPA are confident that you will be satisfied with your new firewood machine. It meets all safety requirements imposed by the European Union and carries the relevant CE label.

1.1 Customer registration

Laitilan Rautarakenne Oy utilises an EXTRANET service where the owners and operators can register their machines. The site contains useful information, such as operating and spare parts manuals.

<https://info.japa.fi/>



FILL IN THE TYPE PLATE INFORMATION AND THE RETAILER'S CONTACT INFORMATION:

Serial number: _____

Year of manufacture: _____

Retailer: _____

Shop: _____

Address: _____

Telephone: _____

1.2 Declaration of conformity

Manufacturer:

Laitilan Rautarakenne Oy
Kusnintie 44
FI-23800 Laitila, Finland
Tel. +358 2857 1200
Fax +358 2857 1201
Web: www.japa.fi

Person responsible for the technical file: Jami Paavola

The declaration applies to the following machines:

JAPA 305 BE	5.6 t	Combustion engine drive
JAPA 305 E	5.6 t	Electrical drive
JAPA 305 TR	5.6 t	Tractor drive
JAPA 305 TRH	4.0/5.6 t	Tractor hydraulics drive

The following directives have been observed in the construction of each machine:

Machinery Safety Directive 2006/42/EC brought into effect through Government Decree 400/2008.

Notified body:

No. 0504
MTT Mittaus ja standardisointi (Vakola)
Vakolantie 55
FI-03400 Vihti, Finland

Laitila 1 January 2012
Laitilan Rautarakenne Oy



Henri Nurminen
Managing Director

japa
Laitilan Rautarakenne Oy
FI-23800, LAITILA, FINLAND

1.3 Purpose of use

JAPA 305 is an efficient, safe and easy-to-use firewood machine for cutting and slitting firewood. The hydraulic 13" chain saw cuts the logs safely and quickly up to a diameter of 30 cm. The blade chain is lubricated with a separate system for supplying blade chain oil. The blade chain only rotates during sawing, which makes working with the machine safer and quieter. The cutting can be freely adjusted between 20 and 45 cm. The splitting activates automatically once the cutting is complete. The machine can be equipped with a blade that splits the logs to four or five pieces. We place great emphasis on the reliability of our products; Japa 305 does not contain any V-belts!



THE FIREWOOD MACHINE IS INTENDED TO BE USED BY ONLY ONE PERSON AT A TIME!

1.4 Instruction and warning labels on the machine



Wear ear protection and protective goggles.



Wear suitable work clothing, gloves and shoes.



Read the user manual before operation and maintenance. Disconnect the power sources before maintenance.



Check the condition and safety equipment of the machine before operation.



Beware of the rotating cutting blade!



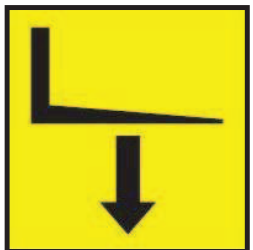
Only one user at a time. Ensure that no other persons are in the working area.



Beware of the splitting blade and the splitting mechanism.



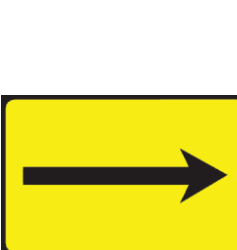
Do not walk under the conveyor! The safety distance is 5 m.



Fork lift point



Maximum allowed tractor hydraulics output.



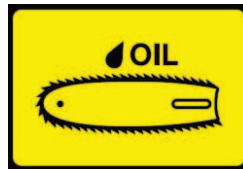
Rotation direction of the electric motor.



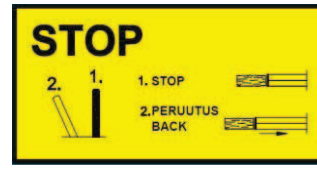
Emergency stop.



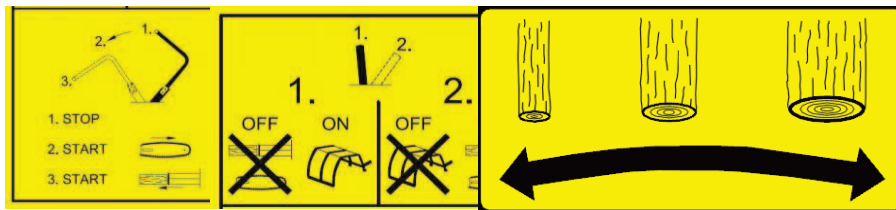
Beware the rotating shaft! The maximum allowed rotation speed and direction of the power take-off shaft.



Tank for blade lubrication oil.



Splitting control (stop-reverse).



Cutting control (cutting-splitting).

Locking of the splitting guard (open-locked). The machine does not operate when the guard is open.

Height adjustment of the splitting axe.



Log length measurement scale.

1.5 The machine's type plate

The machine's type plate is located behind the machine, on the right side of the splitting groove.

The machine's type plate (1)

- Serial number
- Machine type
- Production year and date
- Weight of the machine
- Voltage (machine equipped with an electric motor)
- Maximum rpm of the power take-off
- Maximum hydraulic pressure
- Diameters of the cutting blade/blade opening
- Name and address of manufacturer

Sarjanro		CE
Manufact. Nr		
Malli, Model		
Pvm, Date		
Paino, Weight		
Jännite, Voltage		
R/min		
Hydr.max paine		
Blade		
JAPA - MACHINE LAILAN RAUTARAKENNE OY www.japa.fi, (02) 8571 200, FINLAND		

1.6 Machine models

JAPA 305 BE	5.6 t	Combustion engine drive
JAPA 305 E	5.6 t	Electrical drive
JAPA 305 TR	5.6 t	Tractor drive
JAPA 305 TRH	4.0/5.6 t	Tractor hydraulics drive

Standard equipment in all models:

- 13" hydraulic chain saw and separate chain lubrication system
- Automatically-activated hydraulic splitting with a four-part blade
- Hydraulic output conveyor with a 3.8 m winch and support leg

1.7 Safety instructions

These safety instructions are general in nature. When handling the machine, you must take into account all other relevant instructions concerning health and safety, road traffic regulations regarding transportation, as well as the provisions of the general legislation. Observing the instructions helps to prevent accidents.

The machine may only be used by persons who have familiarised themselves with the manual and the operation of the machine. The operator must not be under the influence of alcohol or narcotic substances. The requirements placed on the users, as well as the required minimum age of 18, must be taken into account.

Carefully study the machine's safety and installation instructions, along with the operation and control functions, before installing and operating the machine.

General provisions:

- The machine has been designed solely for making firewood.
- The machine is intended to be used by one person at a time.
- Ensure that no unauthorised persons are in the working area. The danger zone is 10 m.
- Always check the integrity of any possible power cables.
- Always use approved eye and ear protection.
- Clean and maintain the machine on a regular basis.
- Always stop the machine before maintenance and, if the machine is tractor-driven (TR), disconnect the cardan shaft. If the machine features electrical drive (E), disconnect the power cable.
- Check the operation of the safety equipment. Do not use the machine if part of the mechanism is not functioning properly.
- The machine may only be transported in the transport position. The driver has this responsibility.

Working area:

- Select a flat level and solid working area.
- Keep the area clean and free of obstructions.
- Ensure that the ground or floor is not slippery near the machine.
- Do not use the machine indoors, as the dust will enter your airways or cause a fire hazard.
- Use the machine only with sufficient illumination: daylight is recommended.
- Place the machine in the working position and check the safety devices always before starting the machine.

During operation:

- Take extreme care when cutting knotty or unsuitable wood.
- When sawing incorrectly, the log may spin, causing injury or machine damage.
- Careless sawing or splitting may cause unexpected hazards.
- Ensure that the cardan-PTO shaft is undamaged and attach the shaft guard chain to the machine.
- Avoid unnecessary lifting by using a suitable log mount track.
- Do not lift logs directly onto the input table with loader.



DO NOT LEAVE A RUNNING MACHINE UNSUPERVISED!

Noise level and vibration

JAPA 305's A-weighted sound power level, as detailed in standard EN ISO 3744:2009, is 100.5 dB. In the working area, the average sound power level is 89.5 dB. The weighted power value of the acceleration directed at the arms is 1.19 m/s².

If the machine is driven with a tractor, the tractor may be the dominant source of noise in the working area. Always use sufficient ear protection – earmuffs or plugs. We recommend using a logger's helmet.

1.8 Guarantee terms

Roles

Guarantor

Manufacturer of the JAPA product:
Laitilan Rautarakenne Oy, Kusnintie 44, FI-23800 LAITILA
Tel.: +358-(0)2-857 1200, Fax: +358-2-857 1201, Email: aftersales@japa.fi.

Retailer

The retailer is a company authorised by Laitilan Rautarakenne Oy, which sells and markets JAPA products in its local area. The retailer acts as the recipient in warranty matters regarding JAPA products it has sold to the buyer.

Buyer

The buyer is the person or community that acquires a JAPA product for personal use. The buyer is obliged to report faults within the warranty terms to the retailer and to retain the receipt in order to prove where and when the JAPA product has been purchased. If necessary, the buyer is also obliged to indicate the type plate information to the retailer.

Guarantee's field and area of application

The guarantee is granted for a JAPA product manufactured by Laitilan Rautarakenne Oy.

The guarantee is valid in Finland and all countries included in Laitilan Rautarakenne Oy's export programme.

Period of guarantee validity

The warranty will be effective for one (1) year from the delivery date of the JAPA product in question. The guarantee will become void if the product or its purpose of use – including safety equipment – is modified or changed, or part of the device or equipment is removed from use.

Guarantee coverage

The guarantor is responsible for ensuring that the usability and quality of the JAPA product remain at a normal level throughout the guarantee period. Otherwise, the product is considered to contain a fault. These faults are considered to include, for example, defects in manufacture and construction, along with other such faults and shortcomings discovered during the guarantee period that diminish the product's usability.

Nevertheless, the guarantor is not responsible for any faults where the discrepancy in the product's quality or usability is due to a cause attributable to the buyer. Such causes may include, for example, accidents or using the product in violation to the operating or maintenance instructions provided for the buyer, or in some other incorrect way.

The guarantee does not cover various fluids or regularly-replaceable consumable parts, such as filters, blades, belts and tyres. The guarantee does not cover transport damage or resulting expenses.

Product safety

The fastening mechanisms of parts that affect machine safety as well as factory adjustments that are significant in terms of machine safety have been sealed. If you need to break one of the seals, always contact the machine's retailer or manufacturer. Breaking a seal without an agreement renders the party that broke the seal responsible for any possible consequences. Breaking a seal also voids the machine's factory guarantee.

Procedure in failure situations

The buyer must report the fault to the retailer within a reasonable time frame of observing it.

When reporting a fault, the buyer must present a guarantee certificate, proof of purchase, or other reliable indication as to where and when the product has been purchased. However, this clarification is not required if the place and time of purchase are indicated in a register maintained by the retailer. The retailer will submit a claim for compensation to the guarantor.

The guarantor's obligation to remedy a fault or replace the product with one that is faultless

The guarantor will repair the fault or supply a faultless component within a reasonable time frame from the day on which the retailer reports the fault. The replacement part may also be a component repaired under the guarantee. Guarantee repairs do not extend the guarantee period for the entire product or individual components.

The guarantee covers parts damaged in normal use due to faults in material or manufacture. If the part in question is a hydraulic, electric or power transmission component, the guarantor will require the damaged part to be returned to the factory before the new replacing part is sent to the retailer free-of-charge.

Reasonable expenses caused by repairing a fault

Primarily, the guarantor is entitled to repair the fault if this can be done within a reasonable time in terms of the nature and scope of the fault and in such a way that the repair does not cause expenses or substantial detriment to the buyer.

If the fault cannot be repaired within a reasonable time, the repair work can be conducted at a repair shop recommended by the retailer and authorised by the guarantor, or in some other way (the buyer performs the repairs, for example). In this case, reasonable compensation may be provided for the work, in accordance with the guarantor's agreement.

This reasonable compensation will be primarily provided in the form of product compensation. No discounts will be given on the product. The reasonable costs caused by repairing the fault will be compensated in accordance with the written agreement between the retailer and guarantor. The reception of reasonable compensation for the repair work requires observing the manufacturer's instructions regarding, for example, the guarantor's consent and returning the broken component.

Dispute resolution

Any disputes arising between the guarantor and buyer will primarily be resolved through the retailer. If necessary, negotiations can also be held without the retailer.

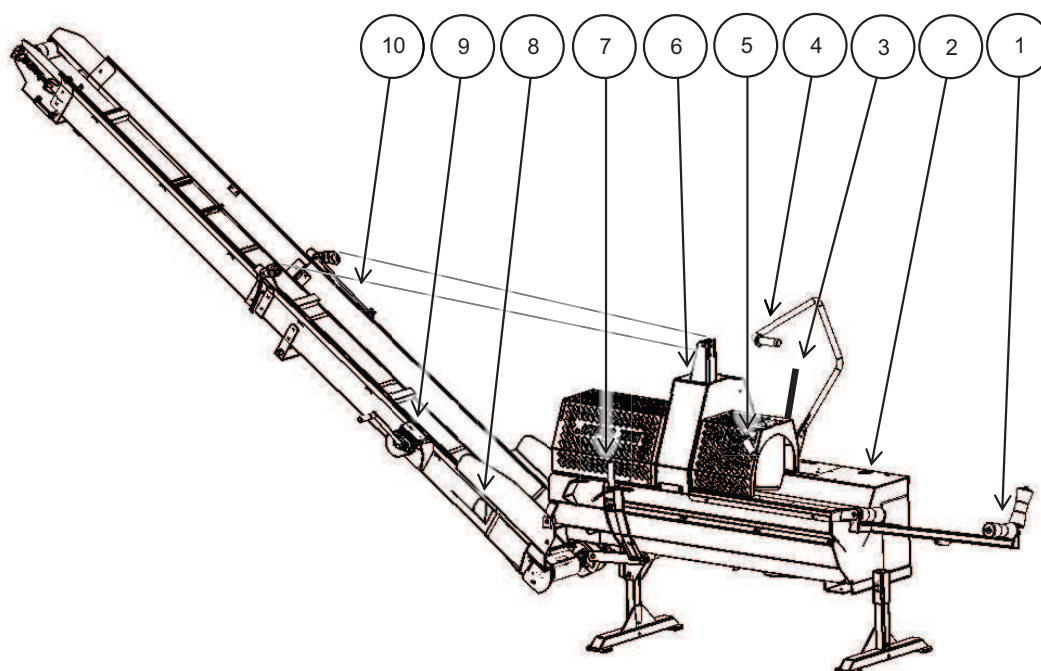
If the dispute cannot be resolved through negotiations between the parties, the consumer buyer may submit the matter to the Consumer Disputes Board for processing. If the dispute is taken to court, it will be processed in the district court of the guarantor's place of business.

2. Installation of the machine

2.1 Delivery inspection

Inspect the machine immediately upon delivery. If the product has sustained damage in transit or it is missing components, immediately contact the transport company and retailer.

2.2 Main components of the machine

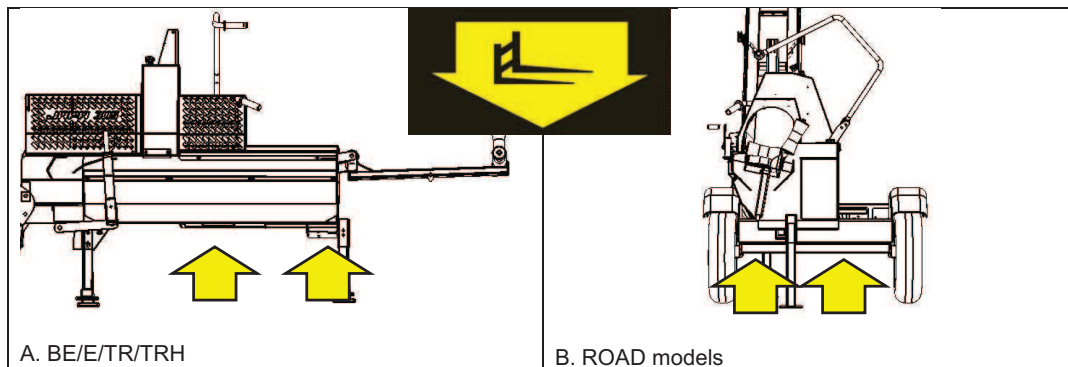


- 1.Extension roller
- 2.Filler opening for hydraulic oil
- 3.Locking switch of the safety mechanism
- 4.Sawing handle
- 5.Wood ~~press~~gripper

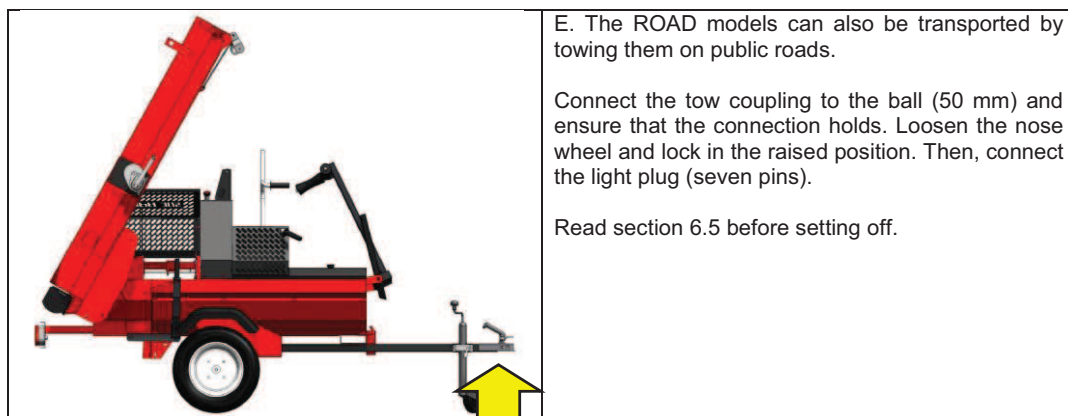
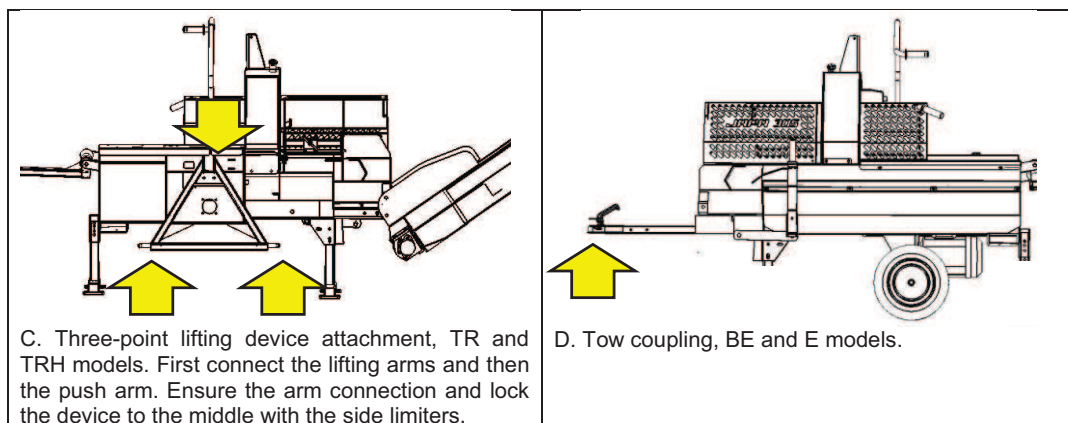
- 6.Filler opening for blade lubrication oil
- 7.Adjustment lever for the splitting blade
- 8.Debris removal opening
- 9.Winch
- 10.Transport support for the belt

2.3 Lifting and transporting the machine

All JAPA 305 firewood machines are equipped with fork lift points with the exception of the ROAD models, where the lifting point is at the front (Figure 2).

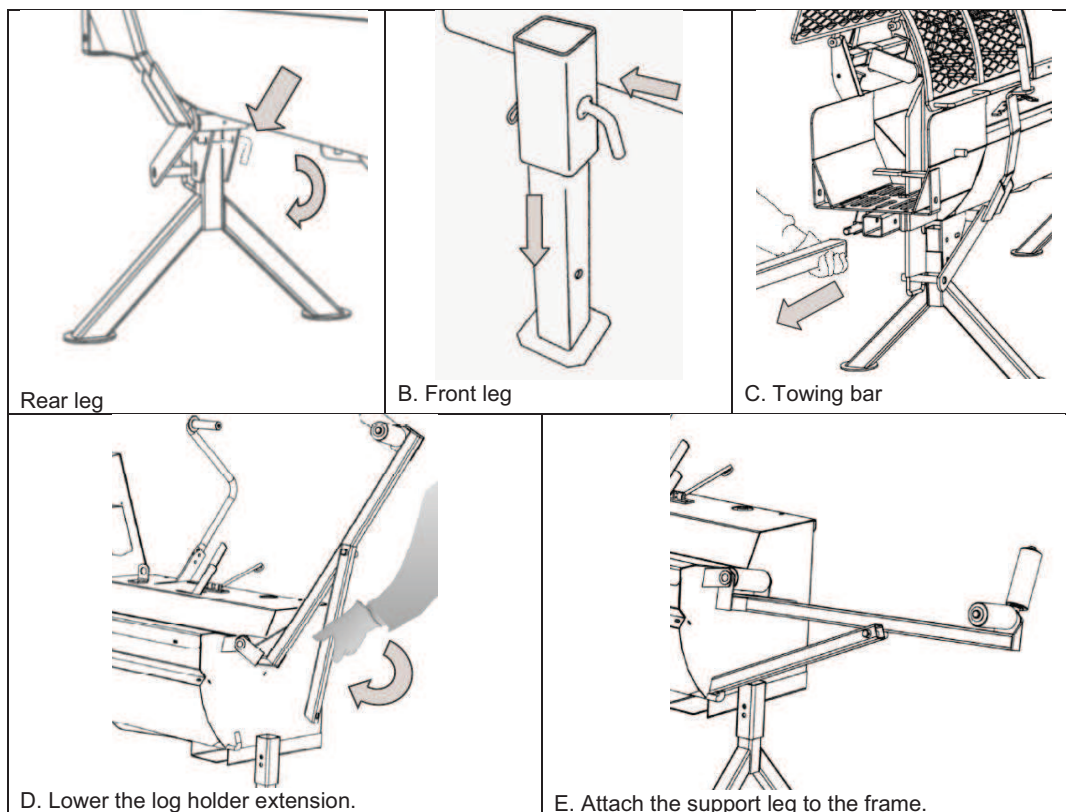


The TR and TRH models are equipped with three-point lifting device attachments CAT1 (2.3.C). The BE and E models feature fixed transport wheels and a tow coupling (2.3.B).

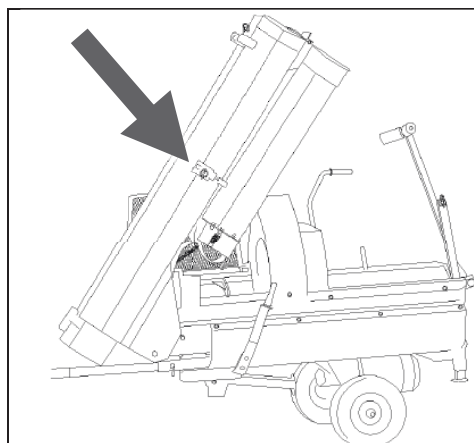


2.4 Working position

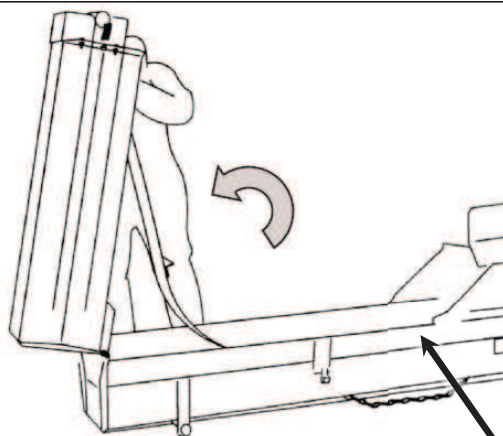
Once you have transported the device to the working location, you can begin placing it in the working position. Ensure that the foundation is flat level and the support legs stand firmly on the ground. Remove the towing bars from the towable models (BE and E) and the light panel under the conveyor from the ROAD models.



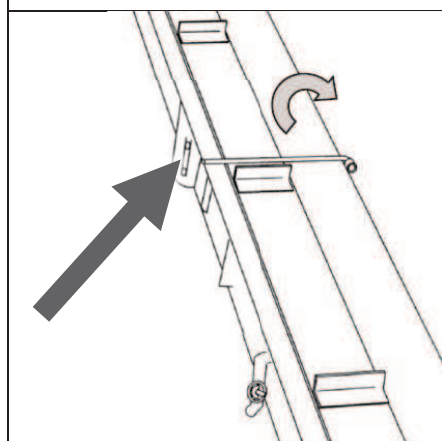
The conveyor is installed in two phases. First, lower the conveyor and fold it open. Then, winch it to the working position.



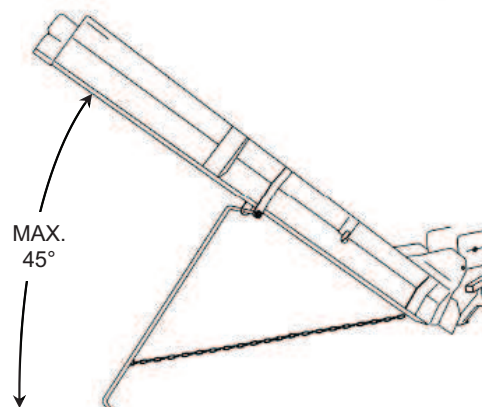
F. Lower the conveyor with the winch.



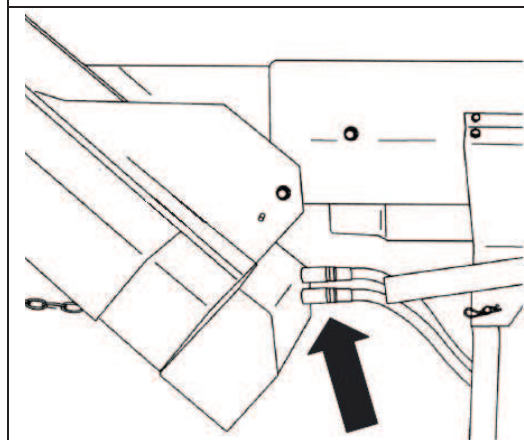
G. Remove the locking pin and fold the conveyor open.



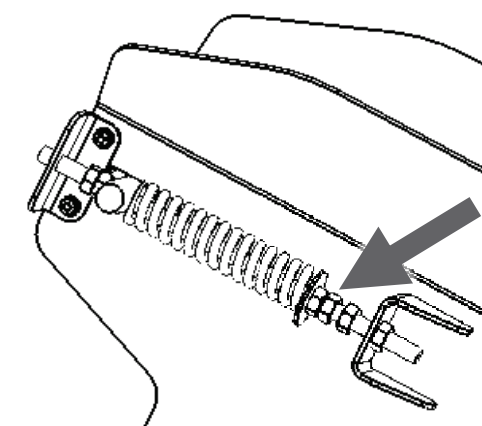
H. Insert the locking pin in the hinge lug and turn the transport support to a longitudinal position.



I. Lift the conveyor to the working position with the winch and support it with the support leg. Note the highest lifting angle.



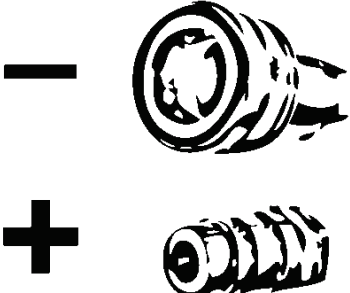
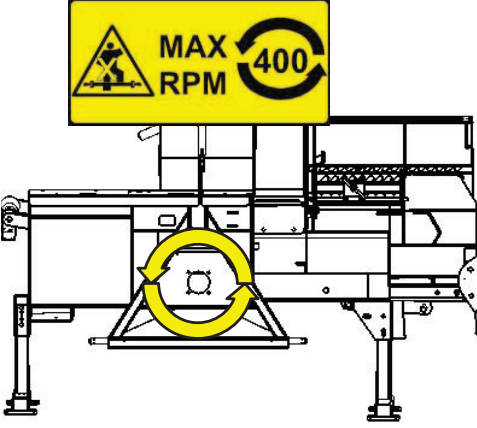
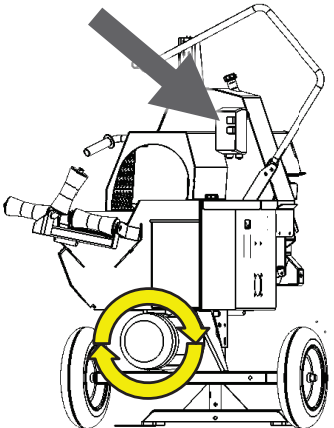
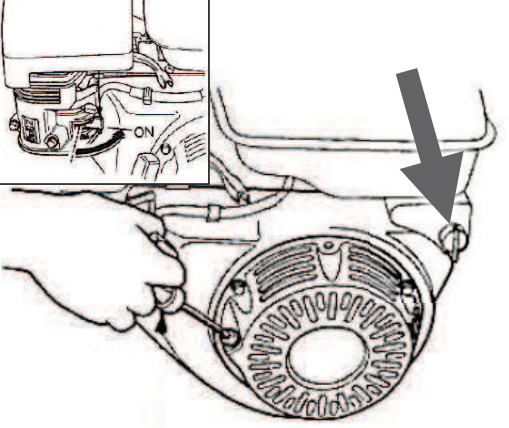
J. Connect the conveyor's hydraulic hoses.



K. If necessary, tighten the conveyor belt (4.4).

2.5 Connecting the power source

The JAPA 305 firewood machine can be equipped with four different power sources. The following includes instructions for preparing them.

MAX. HYDR. 30 L/MIN  A. TRH model: Connect the + connector to the pressure line and the - connector to the available return line. Note the maximum hydraulic output.	 B. TR model: Connect the power take-off shaft. Note the highest rpm 400 1/min.
 C. E model: Electric motor 4kW/400V/16A, connect only to a fault current protected socket. Note the motor's rotation direction and the reverser of the socket.	 D. BE model: Open the fuel valve and turn the power switch to the I position. Pull the starter cable. Read the motor operating instructions for cold starts.



READ THE USER MANUAL BEFORE ACTIVATING THE MACHINE FOR THE FIRST TIME!

2.6 Transport position

The machine can be placed in the transport position by performing the steps in section 2.4 in reverse.

3. Operating the machine

3.1 Acknowledging the hazards and responsibilities involved

Operators of the machine must familiarise themselves with and observe the instructions in the user manual. Incorrect operation of the machine (for example, removing the guards) may compromise the operator's safety. In these cases, the responsibility for the consequences falls on the operator.

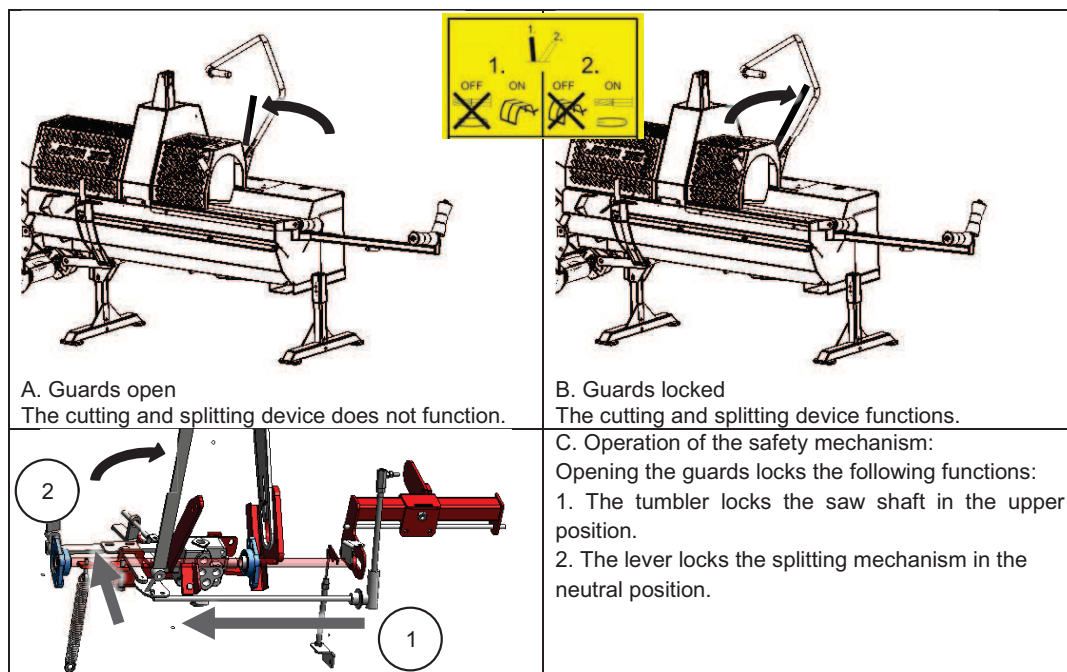
3.2 Before use

Safe use of the machine requires adherence to the instructions provided by the instruction and warning labels. The operator is responsible for the operation and maintenance of the machine and its safety device. Neglecting to maintain the machine or use the safety equipment may void the guarantee. Ensure that you have familiarised yourself with the operation of the machine and its functions.

1. Visually check the machine for damage and oil leaks.
2. Ensure that the machine's power source is correctly connected and firmly in place.
3. Ensure that all of the machine's components and guards are in place.
4. Check the machine's hydraulic and blade oil levels and add oil if necessary.
5. Ensure that no unauthorised persons are in the danger zone and start the machine.
6. Check that the machine's controls work as intended (3.4 and 3.5).
7. Ensure that the cutting and splitting mechanism does not function when the protective mesh is open (3.3).

3.3 Safety mechanism

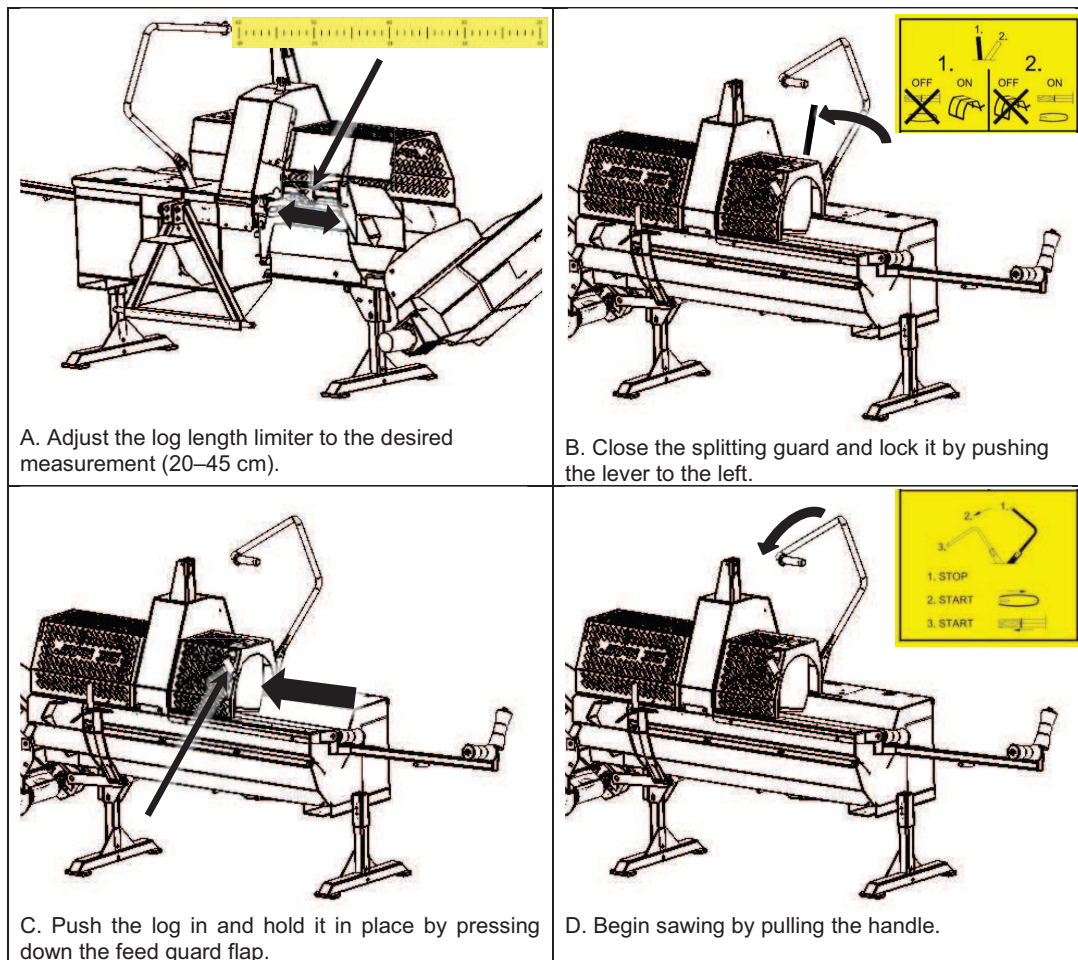
The machine features a safety mechanism that prevents operation when the guard of the splitting section is open. Correspondingly, the guard does not open in the locked position. It must be opened manually.



DO NOT USE THE MACHINE IF THE SAFETY MECHANISM DOES NOT FUNCTION! IF YOU NEED TO ADJUST THE SAFETY MECHANISM, SEE SECTION 1.9 (PRODUCT SAFETY)!

3.4 Cutting device

The Japa 305 firewood machine is equipped with a hydraulic chain saw, which only rotates during sawing. The cutting device is also connected to a safety mechanism, which prevents sawing when the splitting guard is unlocked (3.3). The length of the log to be cut can be adjusted freely between 20 and 45 cm. The splitting motion is activated automatically once the sawing motion ends.



DO NOT PLACE YOUR HAND NEAR THE CUTTING BLADE WHEN THE MACHINE IS RUNNING!



DO NOT HOLD THE SAW HANDLE IN ONE POSITION LEAVING THE CHAIN RUNNING CONSTANTLY!



ENSURE THAT THE LOG BEING SAW IS SUPPORTED AT THE TABLE BACK OF THE TABLE!



DO NOT ATTEMPT TO CUT UNSUPPORTED LOGS, USE THE GUARD FLAP TO GRIP!

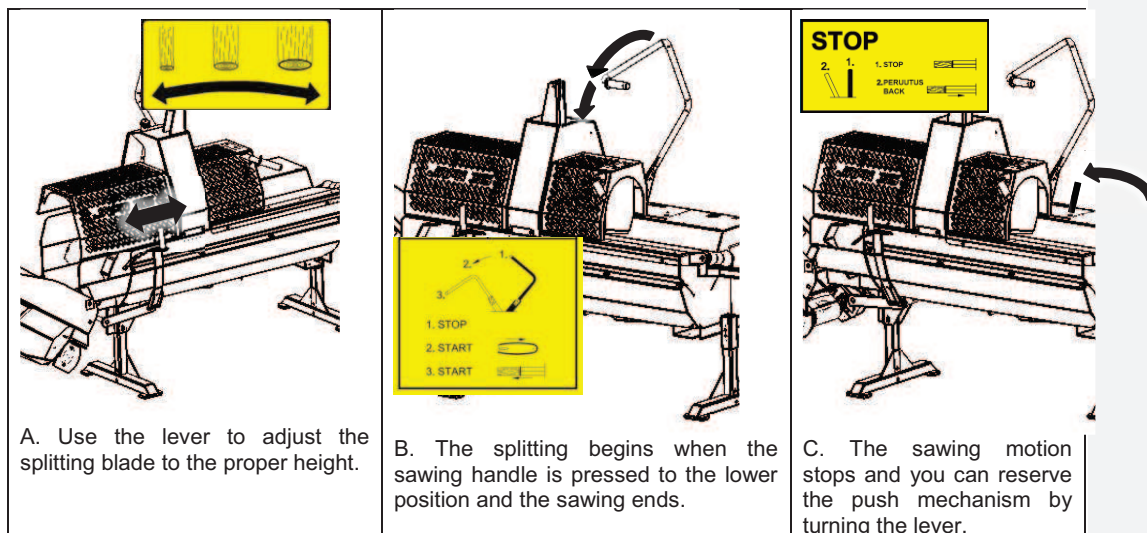
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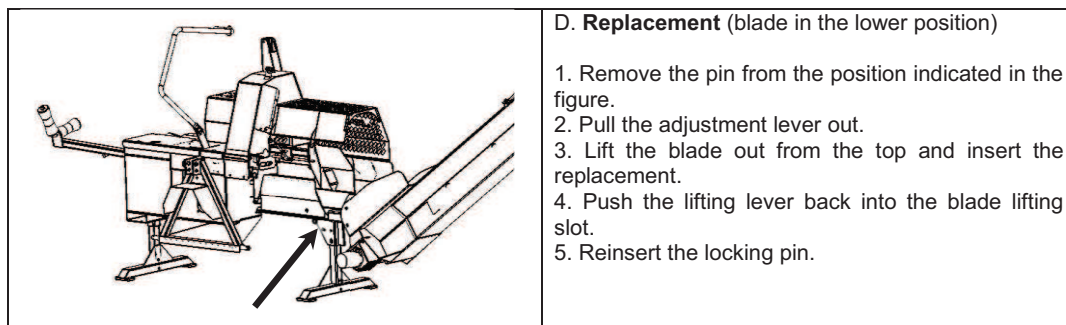
3.5 Splitting device

The Japa 305 firewood machine is equipped with a hydraulic splitting device, which activates automatically after cutting. Moreover, the splitting device does not function if the safety mechanism is disengaged.



3.5.1 Splitting blade

The firewood machine can also be equipped with a blade that splits the log into multiple pieces:



WHEN SPLITTING HARD TIMBER TRY TO SPLIT INTO 2 PIECES ONLY!

Muotoiltu



IF THE LOG TWISTS AFTER SAWING STOP THE PUSHER (3.5.C), OPEN THE GUARD (3.4.B OPPOSITE) AND REALIGN THE LOG IN THE SPLITTING CHUTE!

Muotoiltu



DAMAGE TO THE SPLITTING BLADE FROM NOT ALIGNING LOGS IS NOT COVERED BY WARRANTY!

Muotoiltu

4. Maintenance and troubleshooting, all models

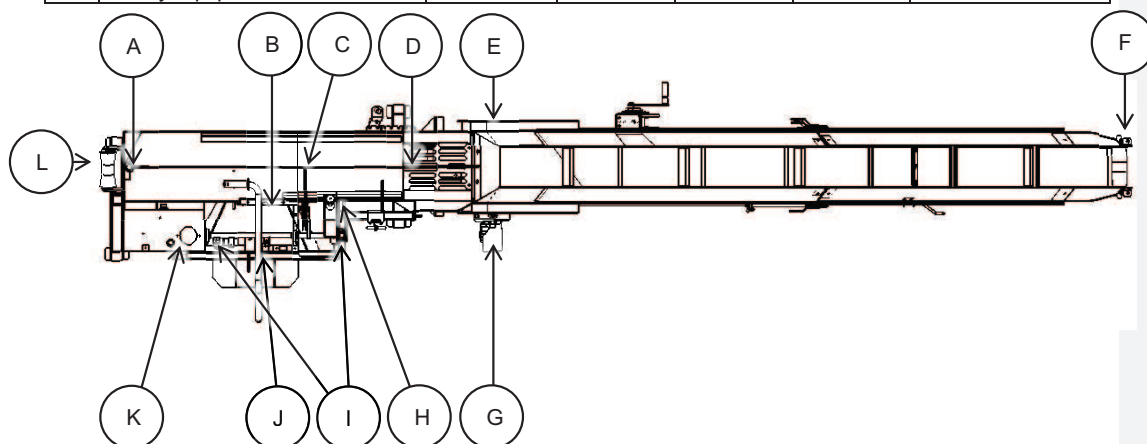
4.1 Maintenance table

Maintenance programmes have been designed for the Japa firewood machines to ensure that their service life is as long as possible. The owner is responsible for maintaining the device. Using the machine carelessly and neglecting maintenance may void the guarantee. The maintenance tasks are itemised as follows:

- 10 h Daily maintenance, to be conducted before use
- 200 h Monthly maintenance, at least once a year
- 1,000 h Annual maintenance, at least every other year

When the machine is used for less than 200 h/year, the 200 h maintenance is performed as the annual maintenance and the 1,000 h maintenance is performed every other year.

	MAINTENANCE TARGET	TASK	INTERVAL 10 h	INTERVA L	INTERVA L	SUBSTANCE/ITE M
				200 h	1,000 h	
A	Splitting device	Clean		x		As necessary
B	Splitting valve	Lubrication		x		Lubrication oil
C	Cutting blade	Check Replace	x			As necessary (96040/96043)
D	Splitting blade	Check Sharpen	x			As necessary (96191)
E	Output conveyor, bearings	Grease Tighten Clean		x		Ball bearing grease
F	Output conveyor, belt			x		
G	Output conveyor, chain coupling			x		
H	Blade lubrication oil	Fill	x			Blade chain oil (1.5 l)
I	Saw shaft, bearings	Grease		x		Ball bearing grease
J	Transmission, oil TR model	Check Replace		x	x	SAE 80W-90 200 ml (0.2 l)
K	Hydraulic oil Normal conditions Oil filter	Check Replace Replace	x		x x	NOT the TR model ISO 46 S (20 l) 94134 / 1 pcs
L	Electrical equipment	Clean	x			
	Machine operation	Check	x			
	Safety equipment	Check	x			



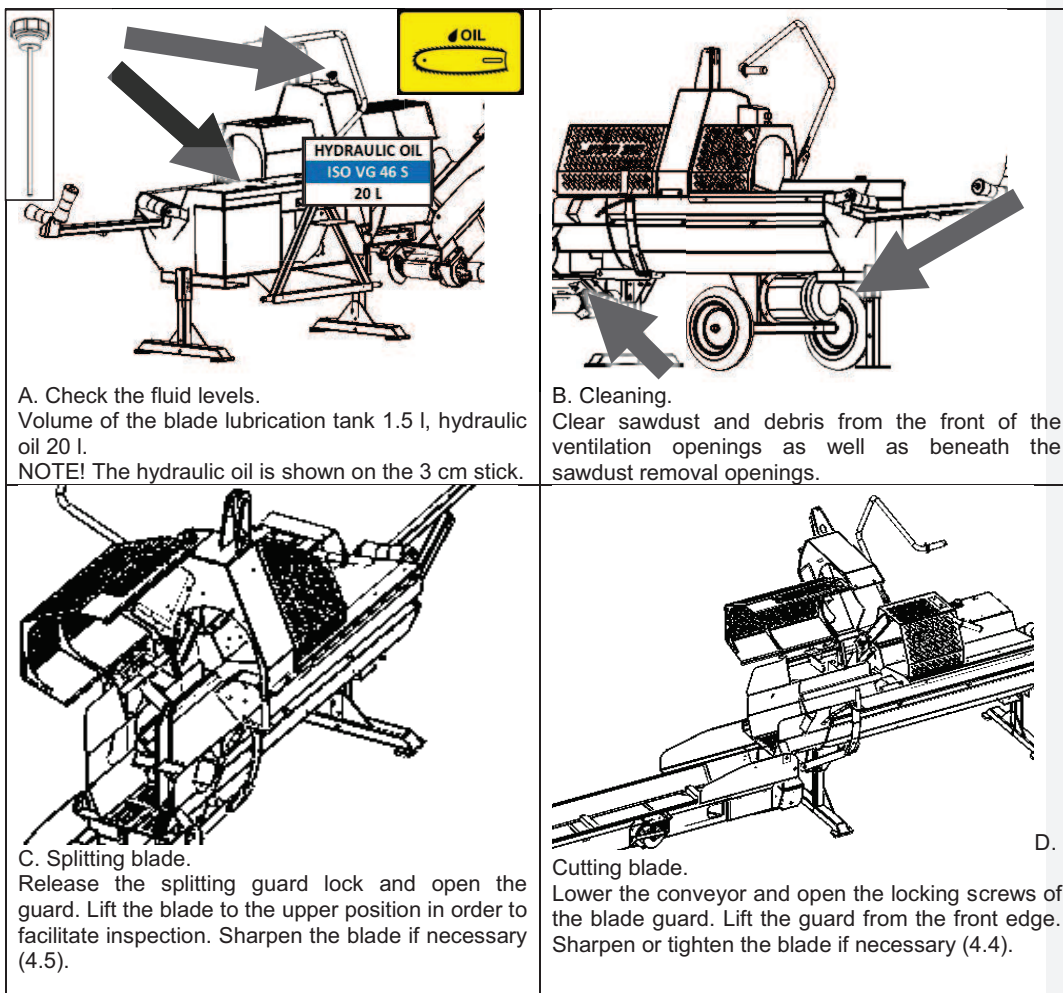
4.2 First maintenance

In order to ensure a long service life for your machine, we recommend replacing the hydraulic oil filter in conjunction with the first 50 h maintenance. This ensures that all impurities are removed from the system.

4.3 Daily maintenance (10 h)

The daily maintenance is intended to be performed between instances of use. The maintenance consists of checking the fluid levels and the cutting and splitting blades, and testing the operation of the machine and safety equipment. Test the operation of the machine and its safety equipment in accordance with section 3.

Always perform these tasks when the machine is inactive and the power source is disconnected.



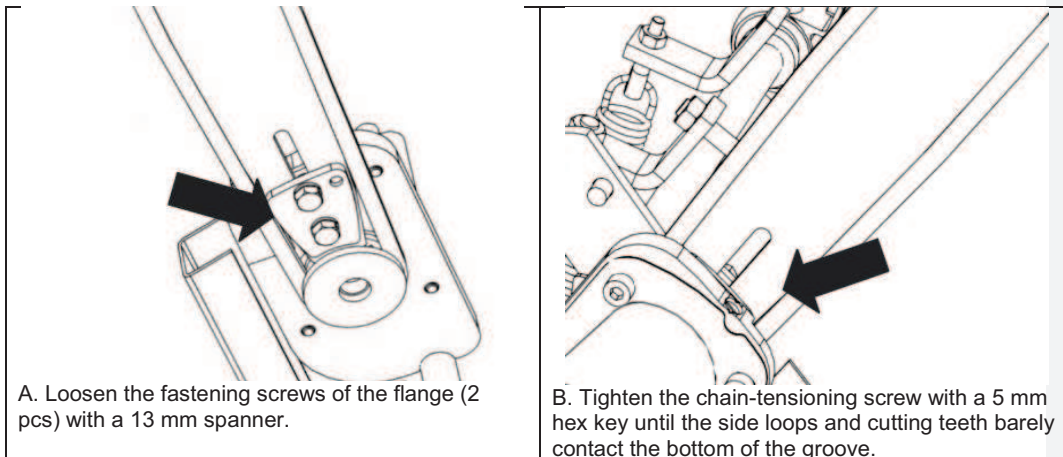
ALWAYS DEACTIVATE THE MACHINE AND DISCONNECT THE POWER SOURCE WHEN PERFORMING MAINTENANCE!



THE TRH MODEL DOES NOT FEATURE ITS OWN HYDRAULICS – THE HYDRAULIC TANK IS EMPTY!

4.3.1 Maintaining the cutting blade

The Japa 305 firewood machine is equipped with a chain saw whose type corresponds with the 13" chain and flange of a traditional chain saw. The pitch is 0.325 with 56 loops. Check the blade on a daily basis and sharpen or replace it as necessary. We recommend turning the flange-bar every time you replace the chain so that the possibly worn guide groove does not damage the new chain.

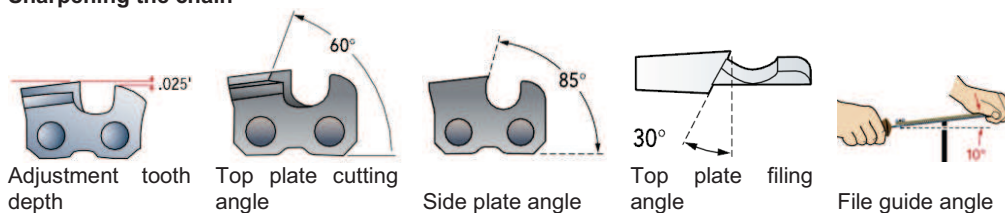


ALWAYS DEACTIVATE THE MACHINE AND DISCONNECT THE POWER SOURCE WHEN PERFORMING MAINTENANCE!



AFTER MAINTENANCE, ALWAYS REATTACH ALL THE GUARDS YOU HAVE REMOVED AND CHECK THE OPERATION OF THE SAFETY DEVICES BEFORE BEGINNING OPERATION!

Sharpening the chain



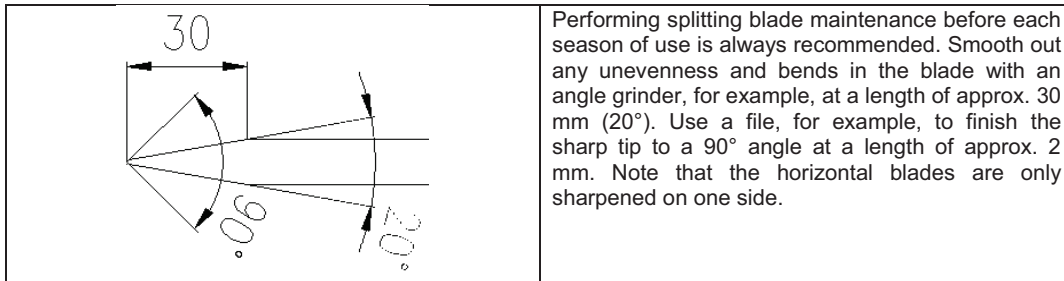
Chain replacement

- When installing a new chain, soak the chain in oil overnight to ensure that the oil penetrates all sections of the chain.
- Do not use a worn drive wheel or flange with a new chain. We recommend that you replace the flange-bar after every other chain, and the chain-wheel/drive sprocket after every third chain.
- Once you have installed the new chain, perform quick pumping motions with the sawing handle without machine running before beginning operation, so that sufficient oil is applied to the chain.
- When starting off, check the chain tension frequently. The new chain tends to initially conform to the flange.
- Saw lightly at first and avoid excessive pressure.



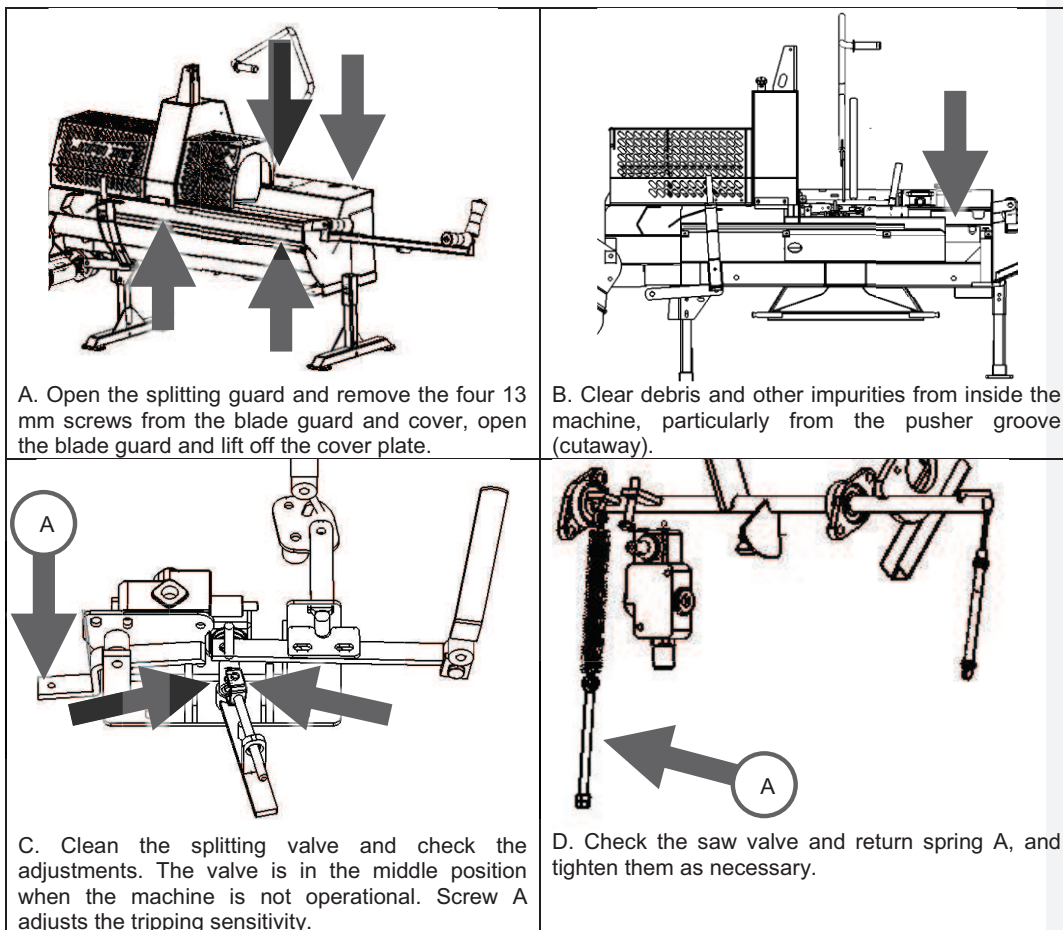
USE PURE MINERAL OILS INTENDED FOR CHAIN LUBRICATION ONLY. DO NOT USE BIO OR WASTE OIL!

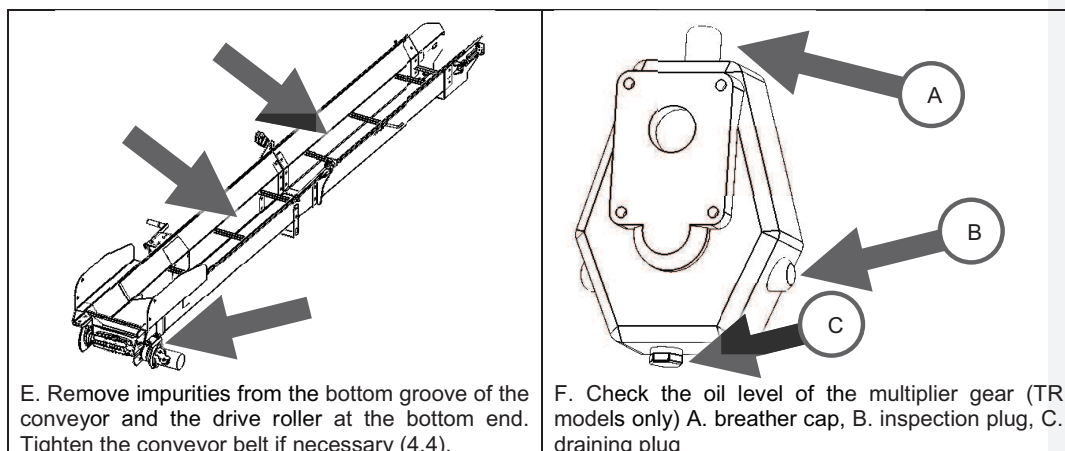
4.3.2 Maintaining the splitting blade



4.4 Monthly maintenance (200 h)

The 200 h maintenance is intended to be performed between long work periods, roughly after every 20–30 days of use. Clean the machine and check the adjustments to prevent problems.





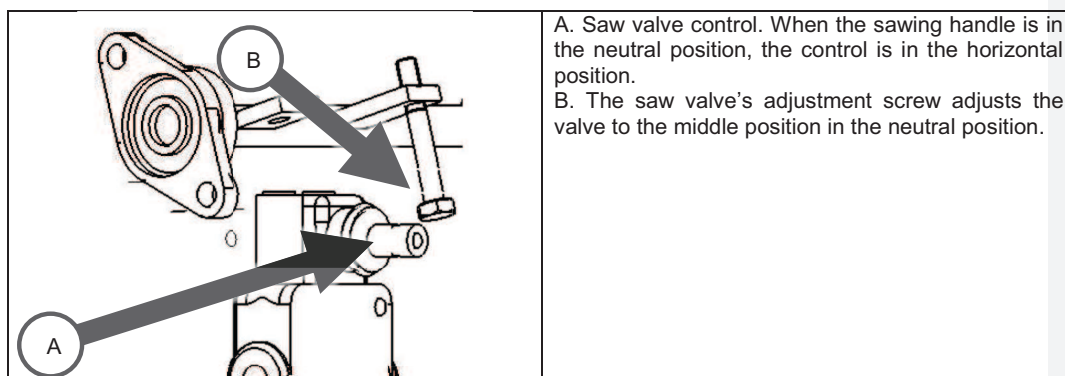
⚠ ALWAYS DEACTIVATE THE MACHINE AND DISCONNECT THE POWER SOURCE WHEN PERFORMING MAINTENANCE!

⚠ PERFORM THE MAINTENANCE PROCEDURES LISTED IN THE COMBUSTION ENGINE'S MANUAL (BE MODELS ONLY)!

⚠ AFTER MAINTENANCE, ALWAYS REATTACH ALL THE GUARDS YOU HAVE REMOVED AND CHECK THE OPERATION OF THE SAFETY DEVICES BEFORE STARTING OPERATION!

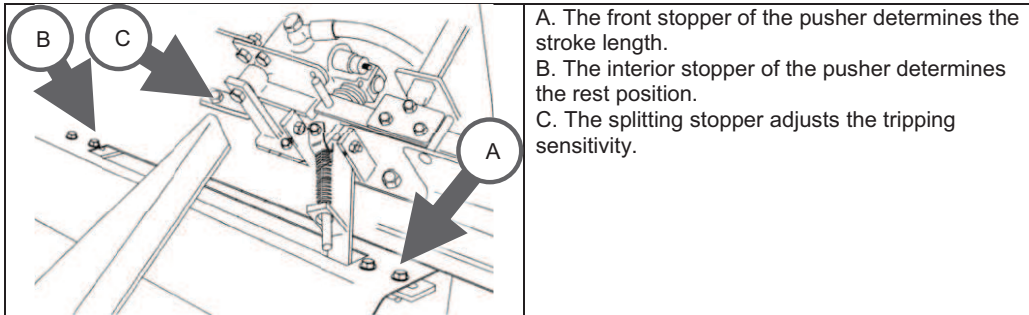
4.4.1 Adjustments of the cutting device

An incorrectly adjusted cutting device may cause hazardous situations as the saw may, for example, keep running or heat the hydraulic oil excessively. Always check the following adjustment in conjunction with the monthly maintenance and tighten the return spring (4.4 D).



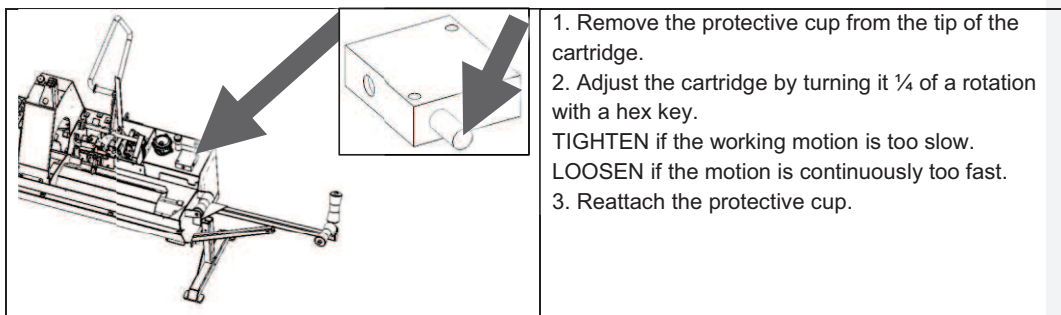
4.4.2 Adjustments of the splitting mechanism

The machine must also be cleaned from the inside in order to ensure the operation of the moving components. Remove the protective plates under the log holder rails and clear the groove and guides of the pusher from any impurities. Always perform these tasks when the machine is inactive and the power source is disconnected.



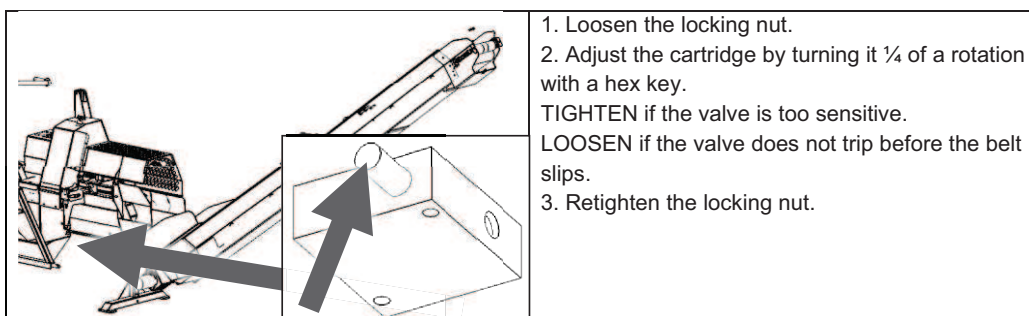
4.4.3 Adjustment of the acceleration valve

The 5.6 t models feature an automatic acceleration valve, which adjusts the power and speed of the splitting cylinder. Without resistance, the working motion proceeds at half power and full speed. If resistance is detected, the cylinder speed slows down and splits at full power.



4.4.4 Adjusting the conveyor's relief valve

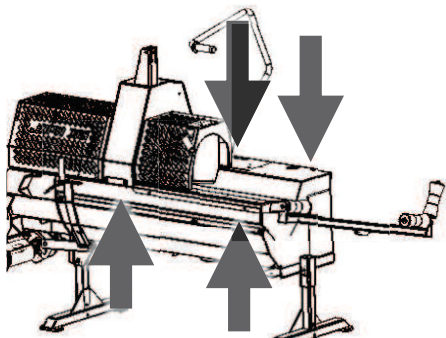
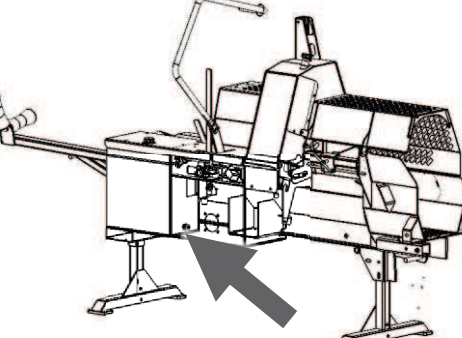
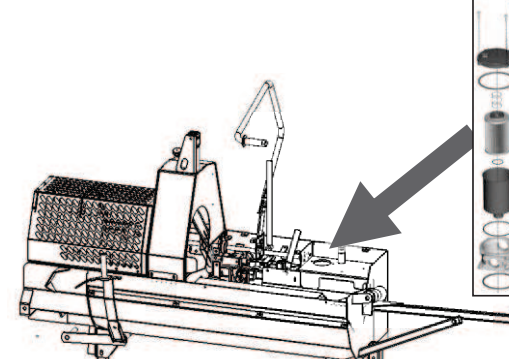
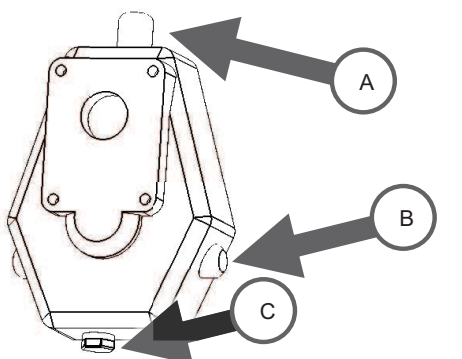
All 305 models are equipped with a hydraulic output conveyor and a relief valve that prevents the conveyor from being damaged if jammed. If the conveyor is overloaded, the valve engages a bypass.



4.5 Annual maintenance (1,000 h)

The 1000 h maintenance is intended to be performed once or twice a year, after approx. every 100–150 days of use. The aim is to go through the entire machine, replace filters and perform adjustments. The 200 h maintenance tasks are also performed at the same time (4.4). Check all hydraulic hoses for leaks or damage, and replace damaged hoses immediately!

TRH models do not require the 1,000 h maintenance since they utilise external hydraulics.

 A. Open the splitting guard and remove the four 13 mm screws from the blade guard and cover, open the blade guard and lift off the cover plate.	 B. Disconnect the suction hose from the side of the hydraulic tank and drain the tank. Reattach the hose when the tank is empty. You can also drain the tank through the filler opening with a suction pump.
 C. Detach the filter cover and remove the old filter cartridge and casing. You can add oil (20 l) when the cartridge is removed. Install the new filter and replace the cover.	 D. Remove bottom plug C and drain the gear. Reinstall the plug and add transmission oil 0.2 l.



ALWAYS DEACTIVATE THE MACHINE AND DISCONNECT THE POWER SOURCE WHEN PERFORMING MAINTENANCE!



THE TRH MODEL DOES NOT FEATURE A HYDRAULIC PUMP – THE HYDRAULIC TANK IS EMPTY!



AFTER MAINTENANCE, ALWAYS REATTACH ALL THE GUARDS YOU HAVE REMOVED AND CHECK THE OPERATION OF THE SAFETY DEVICES BEFORE BEGINNING OPERATION!

4.5.1 Extended maintenance interval

The oil change interval can be extended if the machine is used in excess of 1,000 h a year. The oil must be changed once a year. For intermediate maintenance, it is sufficient to replace the oil filter and, if necessary, add hydraulic/transmission oil.

The filter cartridge must be replaced at least once a year, and the hydraulic oil must be changed at least every other year.

4.5.2 Lubricant table

	HYDRAULIC OIL	TRANSMISSION OIL	BLADE LUBRICATION OIL
GRADE	ISO VG 46 / VG 32	SAE 80W-90 / API GL-4	BLADE CHAIN OIL
AMOUNT	20 litres	200 ml	1.5 litres

4.6 Storage

Perform the daily and monthly maintenance tasks of the firewood machine before storage and oil the blades lightly (for example, spray oil) to prevent rust. Conduct the 200 h maintenance before using the machine again, in order to ensure reliable operation.

Store the machine in a place sheltered from rain.

4.7 Maintenance log

Fill in the maintenance log when you perform the 200 h maintenance.

DATE	HYDR. OIL	HYDR. FILTER	TRANS. OIL	OTHER

5. Troubleshooting

FAULT	POSSIBLE CAUSE	MEASURE
The device does not start.	BE – no fuel BE – power switch in the 0 position E – burnt fuse E – tripped motor protection switch TRH – incorrect hose direction TR – ratchet in the power take-off shaft TR – broken bushing between the gear and pump	Add BE95 Turn the power switch to the 1 position (2.5 D) Replace the fuse Wait for the motor to cool Reverse the hoses (2.5 A) Replace the shaft (2.5 B) Replace the bushing [94131]
The electric motor runs in the wrong direction	Incorrect rotation direction due to the phase sequence	Switch the direction from the socket (2.5 B)
The cutting does not work	The safety lock is disengaged There is no oil or too little of it The chain/flange is damaged Adjustments/operation of the saw valve	Close and lock the splitting guard (3.3 B) Check the level of hydraulic oil (4.3 A) Perform blade maintenance (4.3.1) Check the saw valve (4.4.1)
The saw cuts poorly	The chain/flange is damaged The saw hits the pusher	Perform blade maintenance (4.3.1) Adjust the splitting stopper (4.4.2)
The saw does not stop	The saw shaft does not return to the neutral position The saw valve does not return to the middle position	Tighten/replace the spring [96230] (4.4 D) Check adjustments (4.4.1)
The splitting does not work	There is either no oil or too little There are impurities under/behind the pusher The splitting starts too late The splitting valve's adjustments have changed	Check the level of hydraulic oil (4.3 A) Clean the splitting device (4.3) Adjust the splitting stopper (4.4.2) Adjust the splitting valve (4.4.1)
The motion of the pusher lacks speed or power	There is either no hydraulic oil or too little The oil is too cold The acceleration valve does not change the power	Check the level of hydraulic oil (4.3 A) Allow the machine to run for a few minutes before starting work Adjust the acceleration valve (4.4.3)
The pusher jams at the other end	The relief valve is tripped	Adjust the stroke length (4.4.2) Clean the machine (4.4)
The splitting motion does not stop	The splitting stopper is jammed	Release the stopper (4.4.2)
The log does not split	The splitting blade is incorrectly adjusted The splitter setting exceeds 30 cm The log is jammed on the blade There is a knot in the splitting section, the end is crooked, or the wood is incorrect	Adjust the level of the splitting blade (3.5 A) Remove the log Reverse the pusher, place a smaller piece of wood in the groove, and test the splitting function (3.5) Turn and position the end of the log firmly against and perpendicular to the splitting blade
The oil heats to over 80°C	The power take-off speed is too high Too little/too much oil The cylinder bottoms out The saw valve does not return to the middle position	MAX RPM 400 (2.5 B) Add/remove hydraulic oil (4.3 A) Adjust the splitting device (4.4.2) Check the adjustments (4.4.1)
The log rises up during splitting	The stroke length is too short, which leaves the previous log attached to the axe	Check the stroke length of the piston (4.4.2)
The conveyor belt jams	The belt is loose The logs collide with the conveyor belt The belt is crooked	Tighten the belt (2.4 K) The conveyor angle is too steep (2.4 I) Adjust the top roller of the conveyor
The conveyor does not run	A log has become wedged in the conveyor Wrong rotation direction (E models) The hydraulic hoses of the conveyor are disconnected/loose The relief valve leaks	Remove the wedged log Switch the direction from the socket (2.5 B) Clean and reconnect the quick couplings (2.4 J) Adjust the relief valve (4.4.4)
The cutting and splitting device functions with the guard open	The safety mechanism is damaged	Check and adjust/replace the faulty safety mechanism component (3.3 C)

6. Technical specifications

Cutting	Hydraulic chain saw
Blade flange	13"/1.3mm
Blade chain	56 VL / 0.325" / 1.3 mm
Max cutting diameter	300 mm
Max splitting length	450 mm
Number of splitting pushers	1 kpl
Built-in hydraulic system	Yes (TRH = No)
Electric motor/fuse	4.0 kW / 3 x 16A (E models only)
IP	65
Combustion engine	10 kW (BE models only)
Splitting power/cylinder diameter	4.0 t/40 mm
Splitting power/cylinder diameter	5.6 t/50 mm
Four-piece splitter blade	Standard
Five-piece splitter blade	Accessory
Hydraulic oil tank	20 litres
Blade lubrication oil tank	1.5 litres
Conveyor length	3.8 m
Conveyor belt width	200 mm
Max lifting height	300 cm (45 degrees)
Max weight	490 kg
Max height in working position	305 cm
Max height in transport position	250 cm
Max length in working position	645 cm
Max length in transport position	315 cm
Max depth	133 cm (ROAD models)
Max log diameter	30 cm
Max log length	45 cm
Machine output	2–6 m ³ /h (run)

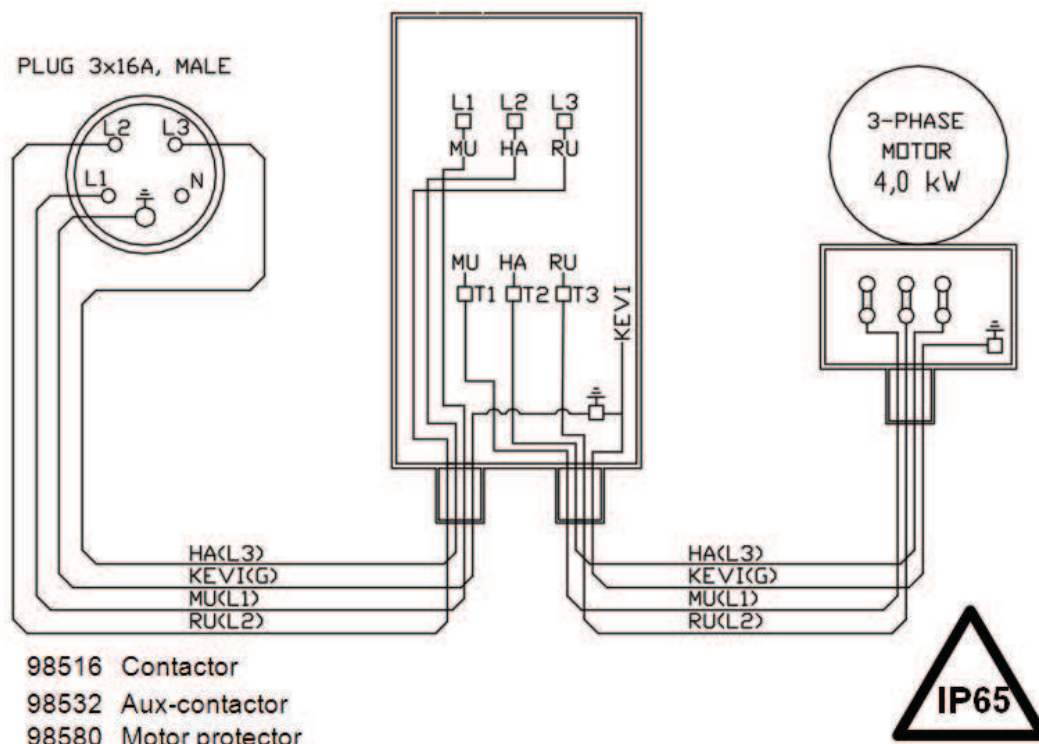
6.1 Technical specifications of the combustion engine (BE models)

Motor type	Honda GX390
Fuel/tank volume	BE95/6.5 l
Motor oil/volume	SAE 10W-40/1.1 l



READ THE MANUAL OF THE COMBUSTION ENGINE BEFORE USE OR MAINTENANCE!

6.2 Electric motor connection diagram (E models)

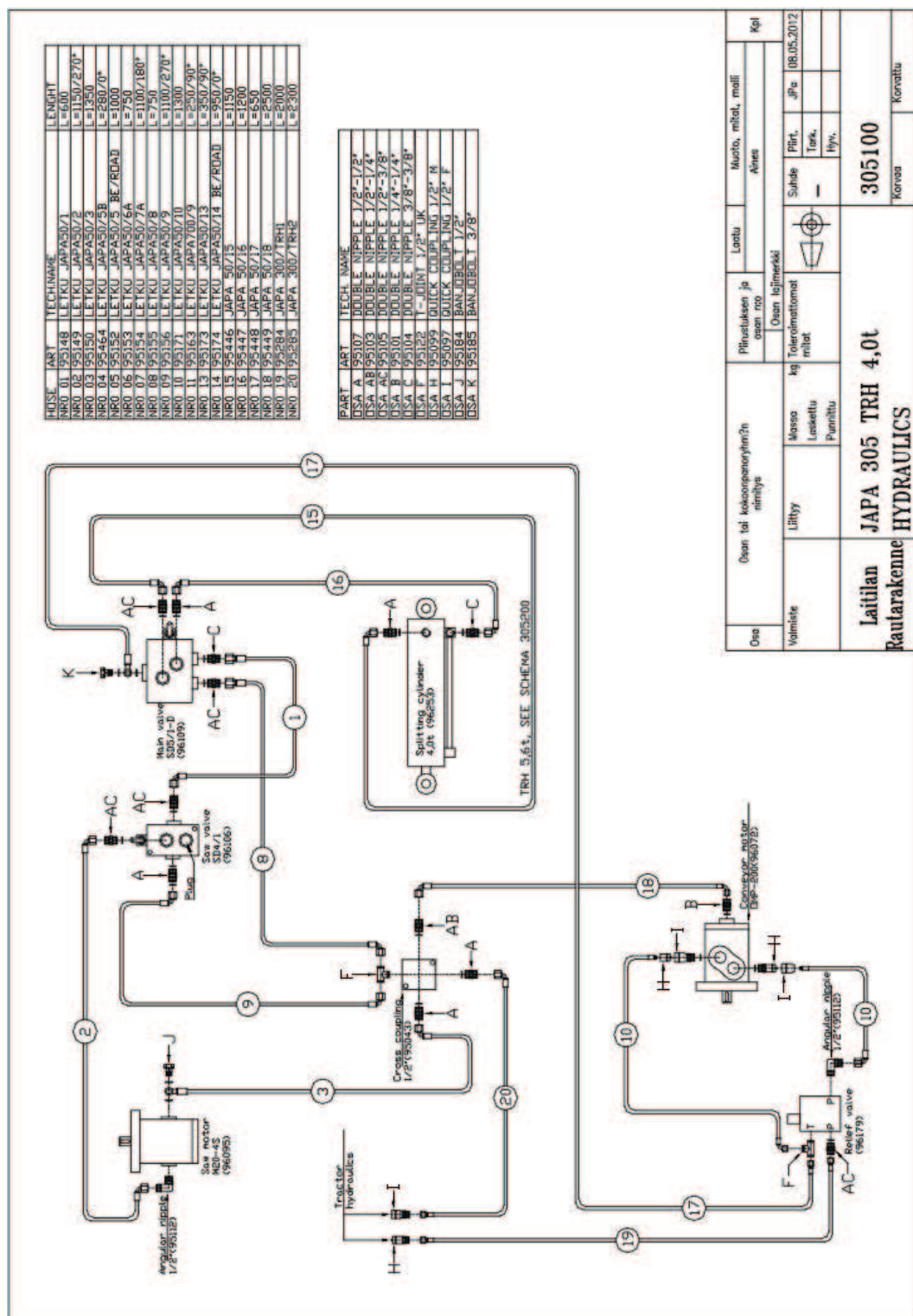


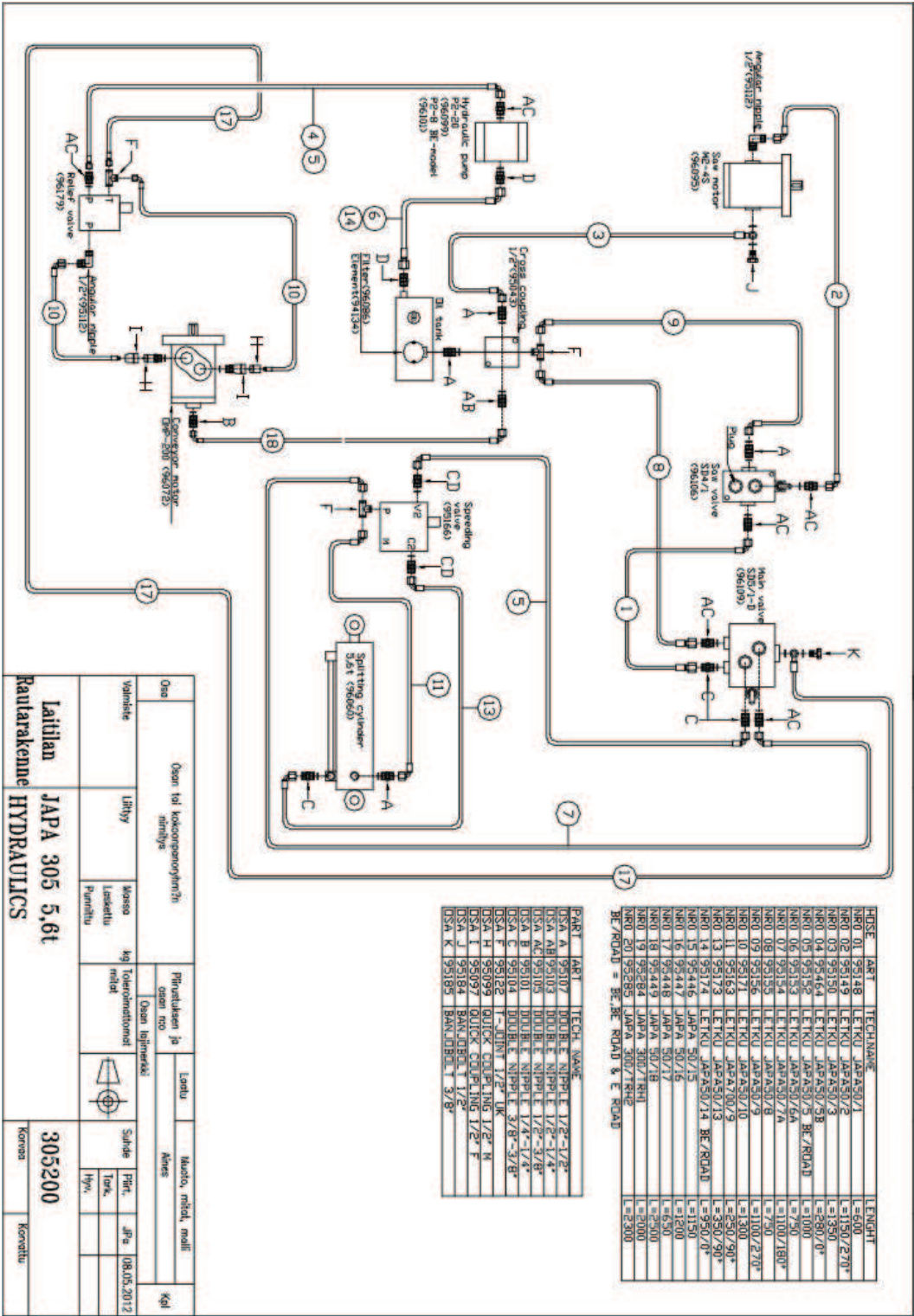
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Laitilan Rautarakenne	JAPA 60E & 100E & 300E ELECTRIC SCHEMA 4kW					98514			
						Korvaa		Korvattu	



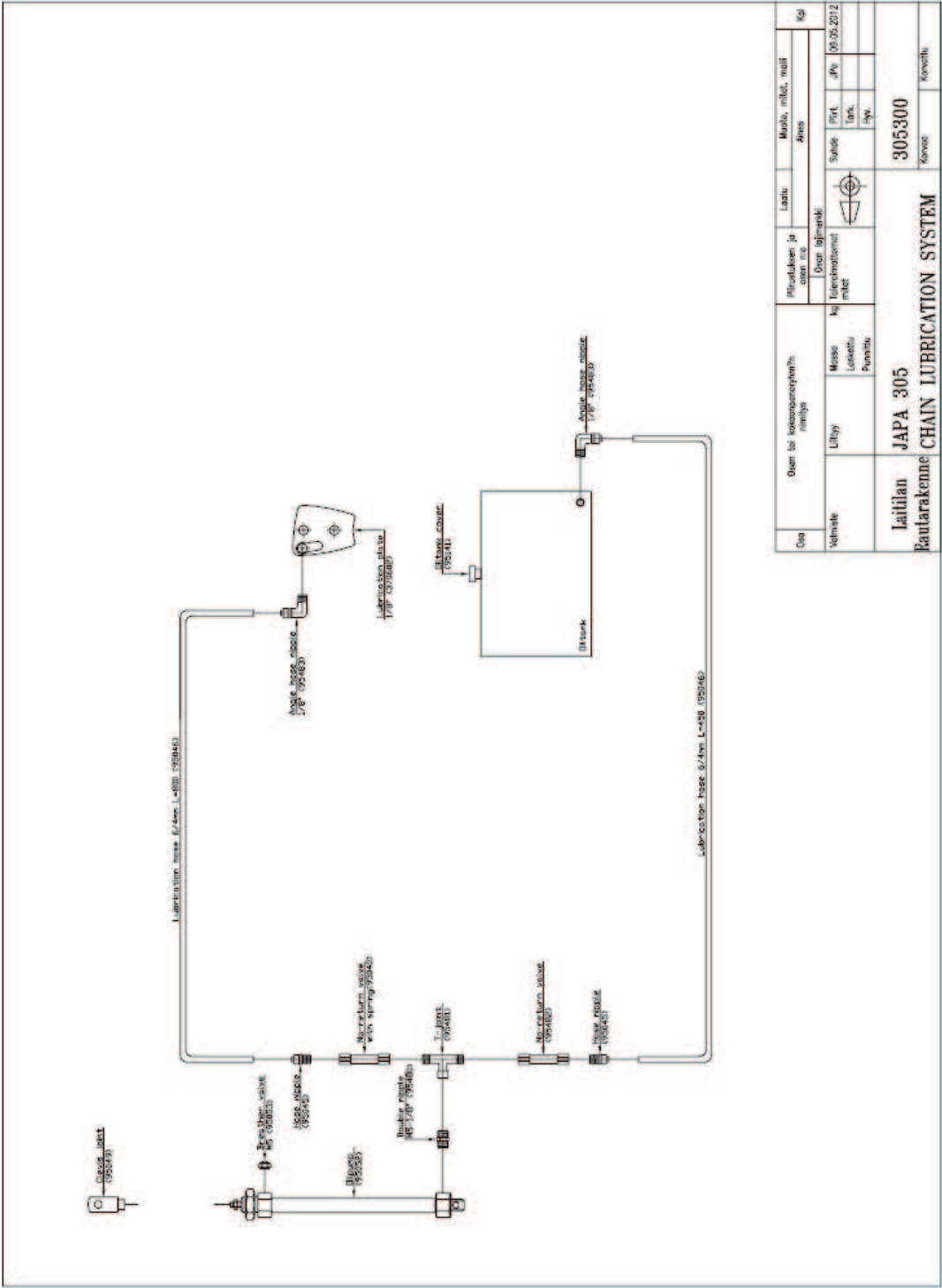
IN THE EVENT OF ELECTRICAL PROBLEMS, CONTACT AN AUTHORISED ELECTRICIAN!

6.3 Hydraulics diagram





6.4 Blade lubrication diagram



6.5 Safety instructions for transport on public roads (ROAD models)

JAPA 305 ROAD models have been designed and constructed for use on public roads. Therefore, all ROAD models feature the appropriate equipment for road traffic, such as lights, reflectors and an underrun guard. The machine also comes with a separate appendix for registration. The retailer can instruct you on how to complete the registration.

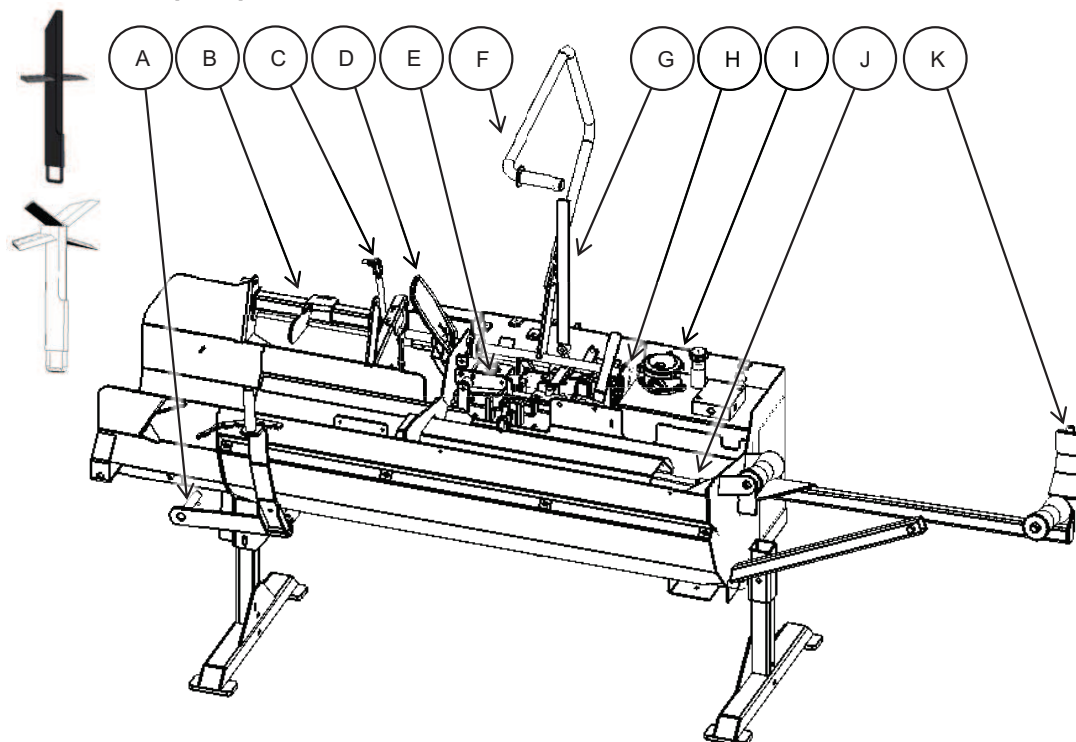
Always check the following before starting your journey:

- Ensure that the tow coupling is attached to the ball and the plug is connected.
- Raise the nose wheel and ensure that it is tightened in such a way that it cannot slip back down during driving.
- Ensure that all lights are functioning: turning signals, brake and rear lights.
- Check the tyre pressure (3.0 bar).
- Fold the conveyor to the transport position and use the winch to tighten it to the raised position. Do not transport anything on top of the machine or in the splitting groove, and clean the machine from debris. Nothing must detach from the assembly that may endanger others on the road.
- If you need to transport the machine on salted roads, we recommend that you wash it immediately after use.
- When the trailer is connected to the towing vehicle, a separate warning triangle must be on hand for the trailer.
- The maximum allowed total mass of the device that does not feature brakes is 540 kg. However, the mass may not exceed half of the mass of the towing vehicle. Note! The car manufacturer may restrict the trailer mass to a lower level. Check the trailer masses of your car from the registration book or user manual.
- The driver must have a class B driving licence if the total weight of the towing vehicle is no higher than 3,500 kg and the total weight of the device being towed is no higher than 750 kg (i.e. the maximum total mass is 4,250 kg), or the total weight of the combination is no higher than 3,500 kg and the total weight of the device (540 kg) does not exceed the net weight of the towing vehicle.
- The driver must have a class BE or ABE driving licence if the combination of the vehicle and trailer/boat trailer exceeds the listed weights.
- Class 01 towable devices (total weight 750 kg or less) do not require periodic inspections concerning eligibility for road traffic.
- Winter tyres do not need to be used on class 01 towable devices.
- The maximum towing speed of a towable device is 80 km/h (60 mph) unless traffic signs impose a lower speed limit.



THE DRIVER OF THE VEHICLE COMBINATION IS ALWAYS RESPONSIBLE FOR THE CONDITION OF THE VEHICLE, PLACING THE TOWED DEVICE IN THE TRANSPORT POSITION, AND THE LAWFUL USE OF THE EQUIPMENT.

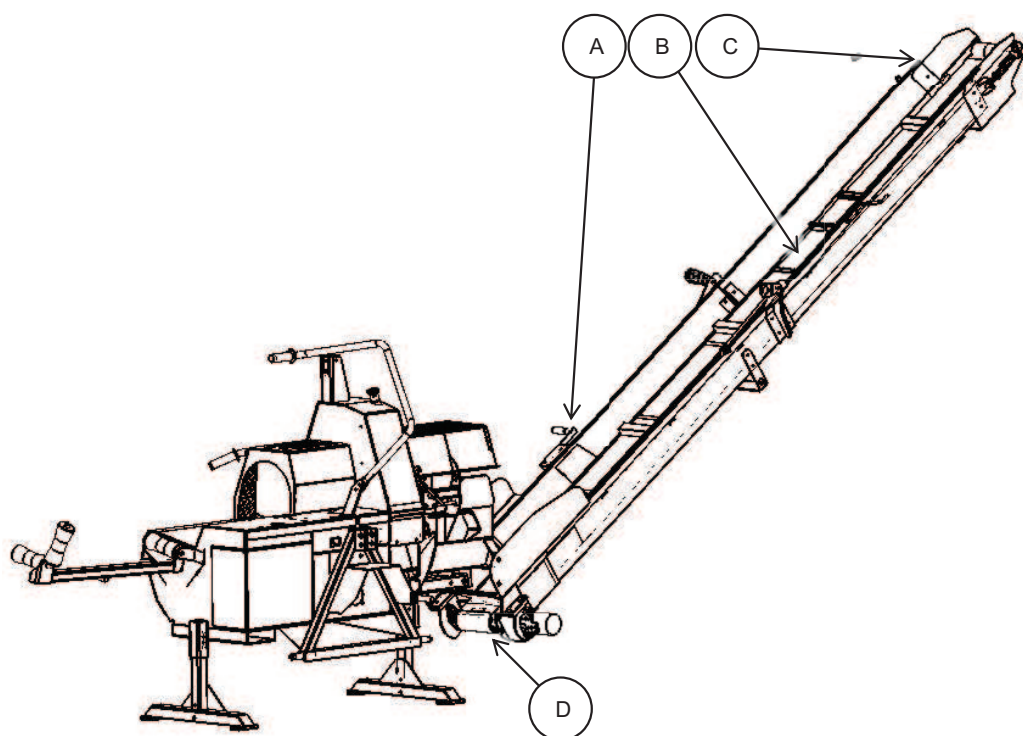
6.6 Spare parts



PART	SPARE PART	NAME
A	505400B	Axe lifter arm
	505600A	Axe lifter handle
	94191	Splitting blade four-piece
	9419794202	Splitting blade six-piece
B	505500C	Log stopper frame
	505516	Stopper plate
C	508371	Splitting guard adjuster rod
D	508100C	Saw shaft
	20008111	Drive wheel
	94138	Saw motor seal kit
	96040	Blade flange 13"
	96043	Blade chain 13"
E	507200B	Splitting stopper
F	20008050	Saw handle
G	507700	Safety mechanism handle
H	96230	Saw shaft return spring
I	94134	Filter cartridge
	94133	Filter seal kit
	95142	Filler cap
J	502100D	Pusher
	94079	Cylinder seal kit 5.6 t
	94074	Cylinder seal kit 4.0 t
K	504500B	Table extension
	504509A	Table extension support

94233

MAINTENANCE PACKAGE 13"



PART	SPARE PART	NAME
A	96181	Winch
B	730060	Conveyor belt 3.5 m, complete
	730018	Belt for the conveyor
	730019	Conveyor plate
	730046	Connection piece
	730023	Bottom rubber
C	730050	Folding roller, assembly
	96135	Bearing
	96016	Tensioning spring
D	730100	Drive roller, assembly
	96072	Drive motor
	96150	Bearing

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